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Nonlocal Fractional Hygro-Photo-Thermoelastic Modeling of Laser-Irradiated Semiconductor Hollow Cylinders for MEMS and Photothermal Sensing Applications

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Abstract

A generalized continuum formulation is established to study the coupled hygro-photo-thermoelastic response of a moisture-laden hollow semiconductor cylinder in a spatiotemporal nonlocal framework. The formulation captures photo-induced carrier diffusion, moisture transport, thermal conduction, and elastic deformation. Heat conduction is governed by Moore–Gibson–Thompson (MGT) theory, thereby overcoming the infinite thermal-wave speed limitation of classical Fourier-based models. To describe memory-dependent transport, the Atangana–Baleanu (AB) fractional derivative with its non-singular Mittag–Leffler (ML) kernel is incorporated into the governing equations. Nonlocal elasticity is also adopted to represent size-dependent behavior and long-range microstructural interactions. For an isotropic hollow semiconductor cylinder under transient laser-pulse excitation, the coupled field equations are solved analytically via normal-mode analysis in the Laplace transform domain. Numerical inversion then yields transient spatial and temporal distributions for temperature, carrier density, moisture concentration, displacement, and stress. **A systematic parametric study reveals that spatiotemporal nonlocality significantly modifies wave propagation characteristics and results in amplified thermoelastic compressive stress amplitudes compared to predictions made by local theories, demonstrating that neglecting nonlocal effects can lead to a significant underestimation of internal stresses.** Fractional memory effects also strongly influence the diffusion of heat, moisture, and photo-generated carriers, producing marked changes in the evolution of the coupled fields. The present model offers a comprehensive analytical framework for complex transport in semiconductor structures under humid and laser-irradiated conditions, with direct relevance to photothermal sensing, microelectromechanical systems (MEMS), and semiconductor device engineering.

Keywords: Hygro-photo-thermoelasticity; MGT theory; Spatiotemporal nonlocality; AB fractional derivative; Laser pulse excitation; Carrier diffusion; Moisture-affected transport; Photothermal sensors.

1. Introduction

The escalating complexity of modern micro- and nano-scale devices has driven a fundamental transformation in semiconductor physics toward modeling frameworks capable of integrating thermal, mechanical, electronic, and environmental interactions with high fidelity. Among the most consequential developments in this domain is the study of photo-thermoelastic phenomena, which has assumed a central role in applications spanning photoacoustic microscopy, high-resolution thermal imaging, and non-destructive semiconductor diagnostics. When a semiconductor medium is subjected to laser illumination, the resulting photoexcited carriers generate localized thermal gradients and mechanical strains, giving rise to the propagation of coupled thermoelastic and plasma waves. These dynamic wave interactions are profoundly governed by the material's underlying microstructure; at micro- and nanoscales, such scale-dependent features render classical continuum descriptions inadequate and necessitate the implementation of nonlocal elasticity. This framework provides a physically consistent description of material behavior by accounting for long-range interatomic forces that are neglected in conventional theories (Mandelis [1]; Almond and Patel [2]; Nikolić and Todorović [3]; Mandelis and Michaelian [4]).

The synergy between photothermal and electronic excitations in semiconductors induces significant deformation of the crystal lattice through the combined contributions of elastic, plasma, and thermal fields. Although this interplay has been extensively documented, the majority of existing studies rely on the foundational assumptions of classical elasticity. Nonlocal theory, originally introduced to incorporate long-range interatomic interactions, offers a superior analytical framework for characterizing these phenomena in small-scale semiconductor systems. In environments where conventional continuum models fail to capture essential size effects, it becomes imperative to examine photoexcited carrier dynamics and thermal diffusion within a nonlocal framework. Such an approach ensures more realistic predictions of wave propagation characteristics and stress distributions (McDonald and Wetsel [5]; Jackson and Amer [6]; Stearns and Kino [7]; Todorović [8–10]).

To address the limitations of classical heat conduction, numerous studies have extended the field of thermoelasticity to incorporate generalized heat conduction, viscoelasticity, and various microstructural effects. Sharma [11] and Sharma et al. [12] investigated the complexities of generalized thermoelastic diffusion and viscoelastic models, while Marin et al. [13] extended the reach of Saint-Venant's principle by introducing micropolar thermoelastic diffusion. Analogous methodologies have been successfully applied to specialized problems in bioheat transfer (Sharma and Sharma [14]). Despite these advances, the majority of such investigations remain confined to a local elasticity framework. The integration of nonlocality is essential for nanoscale systems, where constitutive stress–strain relations are strongly influenced by the material's internal characteristic length (Hobiny and Abbas [15]; Marin et al. [16]).

In parallel, fractional calculus and phase-lag models have been widely employed to address heat conduction in semiconductors, emphasizing the critical importance of memory and relaxation effects. Sharma and Kumar [17–19] applied generalized and fractional approaches to model photothermoelastic deformation under dynamic loading conditions. While these methods successfully account for temporal nonlocality in thermal transport, they often fail to fully capture spatial nonlocality in the mechanical field. Nonlocal elasticity thus serves as a necessary complement to these models by addressing spatial scale effects in stress distribution and wave dispersion, thereby ensuring a more complete and accurate representation of semiconductor responses (Sharma and Khator [20, 21]).

At the nanoscale, the operational stability of semiconductors is further complicated by environmental factors such as moisture diffusion, porosity, and external magnetic fields (El-Sapa et al. [22]; Sherief et al. [23]). Building upon the established frameworks of photothermal

and thermoelastic theories, the integration of environmental variables—specifically moisture—has emerged as a critical frontier in materials science. The mathematical foundation for this coupling was significantly advanced by Szekeres [24, 25, 27], who identified a deep physical analogy between heat and moisture transport, establishing the theory of cross-coupled diffusion in which temperature gradients drive moisture flow and vice versa. This hygrothermal perspective is essential for understanding materials subjected to variable environmental conditions, as further evidenced by the coupled models developed for porous structures such as concrete by Gasch et al. [26]. The resulting discipline of hygrothermoelasticity [28, 29] provides the necessary tools to analyze how the simultaneous diffusion of heat and moisture alters the mechanical stress state of a body. This framework has recently been applied to complex anisotropic media, including the orthotropic piezo-hydro-thermo-elastic systems studied by Yadav et al. [30] and the investigation of solid cylinders under hyperbolic two-temperature theory by Kaur and Singh [31].

Research in the 2020s has shifted toward ultra-complex multi-physical models operating at micro- and nanoscales. Alshehri and Lotfy [32] introduced a photo-thermoacoustic-diffusion model incorporating chemical potential into hydro-poroelastic nanocomposites, representing a significant leap in modeling semiconductor behavior in chemically active environments. Concurrently, Kumar [33, 34, 36, 37] and Kaur [35] have extensively utilized modified couple stress theory and memory-dependent derivatives to analyze the vibration and deformation of thick circular plates. These studies highlight that at small scales, traditional local elasticity theories fail to capture the size effect, necessitating the use of nonlocal parameters to accurately describe the internal interactions of the crystal lattice. To solve the resulting high-order governing equations, the numerical inversion of Laplace transforms, as detailed by Honig and Hirdes [38], remains a vital computational bridge between the frequency and time domains.

The most recent breakthroughs, pioneered largely by Abouelregal [39–41], revolve around the synthesis of MGT theory with nonlocal and fractional-order calculus. Researchers in this area have applied such advanced frameworks to a range of cutting-edge problems, including rotating magnetized semiconductors under ultrafast laser excitation [42] and the size-dependent thermoelastic analysis of functionally graded materials [43]. Further developments include memory-enhanced MGT formulations for nanoscale materials in magnetic fields [44] and the use of nonlocal fractional MGT–non-Fourier models to analyze vibrations in rotating nanobeams and semiconductors with spherical cavities [45, 46]. Nevertheless, the combined consideration of carrier density, moisture diffusion, and nonlocal elasticity in photothermoelastic hollow cylinders has not yet been reported in the literature.

Fractional derivatives extend classical derivatives to non-integer orders, forming a fundamental component of fractional calculus for measuring rates of change. Commonly defined through integral operators [47, 48], the Riemann–Liouville and Caputo fractional derivatives have been the most prevalent, with their selection depending on the specific requirements of the problem under investigation. These derivatives have been widely applied in physics, engineering, and biology, effectively modeling complex dynamics and memory-dependent phenomena such as anomalous diffusion and viscoelasticity [49, 50]. Fractional derivatives with non-singular kernels further generalize classical derivatives for non-integer orders. Unlike classical derivatives, which capture the rate of change at a single point, these operators describe function evolution across time or space, incorporating nonlocal effects. In fields such as physics, biology, and engineering, non-singular kernels, such as exponential or ML functions, play a critical role in accurately modeling complex systems. The choice of kernel depends on the application and is selected to ensure well-defined mathematical properties and practical relevance [51]. Caputo and Fabrizio [52] introduced a fractional derivative with an exponential kernel to address the singularity issues inherent in Riemann–Liouville and Caputo derivatives. However, their operator is nonlocal, and its associated integral does not qualify as

a true fractional integral. Atangana and Baleanu [53] overcame these limitations by proposing fractional derivatives based on the ML function, offering improved locality and mathematical rigor. A unified framework for all non-singular kernel fractional derivative operators was subsequently established, as detailed in [54].

While significant progress has been made in generalized thermoelasticity, the coupling of carrier density, moisture diffusion, and thermal fields in cylindrical semiconductors has already been established within the MGT framework, notably by Singh et al. [46] for a solid cylinder under moisture and thermal diffusivity, and by Kaur et al. [55] for a solid cylinder with hyperbolic two-temperature effects. Both studies, however, adopt classical local elasticity and integer-order or Caputo-type time derivatives. They therefore cannot capture the size-dependent effects, long-range microstructural interactions, and non-singular memory effects that govern transport and deformation at micro- and nanoscales. The genuine contribution of the present work is the extension of this established coupled framework in three specific directions: (i) the incorporation of Eringen-type nonlocal elasticity, which introduces both spatial and temporal scale dependence absent from [46] and [55]; (ii) the adoption of the AB fractional derivative with a non-singular ML kernel, which replaces the integer-order derivatives used in prior MGT semiconductor studies with a memory operator that is simultaneously non-singular and non-local; and (iii) the treatment of a hollow-cylinder geometry with distinct inner and outer boundary conditions, which introduces wave-reflection and interference phenomena that do not arise in solid-cylinder formulations. The present study therefore does not claim a wholly new physical coupling, but rather a mathematically consistent generalization of an existing coupling to the nonlocal, fractional, and hollow-geometry regimes that are essential for the accurate modeling of modern micro- and nanoscale semiconductor devices..

Building on the coupled hygro-photo-thermoelastic framework of Singh et al. [46] and Kaur et al. [55], the novelty of this work resides in the systematic generalization of that framework to include spatiotemporal nonlocal elasticity and AB fractional memory. Specifically, the present formulation: (i) retains the MGT heat conduction equation to ensure finite-speed thermal and moisture wave propagation, as in [46] and [55]; (ii) augments the constitutive and momentum equations with a KG-type nonlocal operator that introduces both spatial scale dependence and temporal relaxation absent from prior cylindrical models; and (iii) replaces the integer-order time derivatives with AB fractional derivatives, yielding a non-singular memory kernel that produces a measurably different damping hierarchy. The combined effect of these three extensions is quantified throughout Section 6 and validated against the classical limit.

The central problem examined here is the transient response of a hollow cylinder whose **outer surface is irradiated by a laser pulse** while its **inner surface** is simultaneously subjected to a **compressive step pressure**, with all field variables required to satisfy the prescribed boundary conditions at both surfaces. The analytical solution, obtained via the normal-mode method, reveals how spatiotemporal nonlocality alters wave dispersion and attenuates thermal shocks. The practical implications of this work are significant: the findings provide a critical guide for the structural optimization of high-precision microelectromechanical systems, moisture-sensitive photothermal sensors, and semiconductor-based actuators. By incorporating moisture and photoexcitation into a unified nonlocal framework, this model enables more resilient designs capable of maintaining high operational reliability in humid, high-frequency operating environments.

2. Basic equations

The three theoretical frameworks adopted here operate at different levels of the coupled system and address distinct physical mechanisms. Eringen nonlocal elasticity modifies the

constitutive and momentum equations by introducing spatial nonlocality through the operator $(1 - \xi^2 \nabla^2)$, thereby capturing long-range interatomic forces and size-dependent effects. MGT heat conduction modifies the energy equation, replacing the parabolic Fourier law with a hyperbolic formulation that ensures finite-speed thermal wave propagation. The AB fractional derivative modifies the time derivatives in the heat and moisture equations, introducing a non-singular ML kernel that captures memory-dependent, sub-diffusive transport.

The three operators are therefore orthogonal in their physical action: Eringen governs spatial scale, MGT governs finite-speed thermal propagation, and AB governs temporal memory. Because they act on different terms of the coupled system, no mathematical contradiction arises, and they are physically complementary, since a semiconductor at micro- and nanoscales simultaneously exhibits all three effects. The formulation satisfies three consistency checks: as $\xi \rightarrow 0$, classical local elasticity is recovered; as $\tau_q \rightarrow 0$ and $K^* \rightarrow 0$, the Fourier law is recovered; and as $\alpha \rightarrow 1$, the integer-order derivative is recovered. When all three limits are taken simultaneously, the system collapses to classical coupled thermoelasticity, as verified in the quantitative validation. No double-counting occurs: τ_q represents a single thermal relaxation time, whereas α governs the spectrum of memory effects, and ξ and α act on independent variables, space and time. Building upon the foundational works of Todorović [8–10], Szekeres [24, 25], and Singh et al. [46], the governing equations for an isotropic, moisture-affected photo-thermoelastic medium are developed under the assumptions of no external body forces, extrinsic heat sources, or exogenous moisture sources.

To accurately reflect scale-dependent phenomena observed at micro- and nanoscale dimensions, Eringen's nonlocal elasticity theory is integrated into the constitutive law. By introducing an internal characteristic length parameter, this approach transcends the limitations of classical local continuum mechanics, allowing for the inclusion of long-range atomic interactions. Consequently, the generalized nonlocal constitutive equation is formulated as follows [44]:

$$\left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) t_{ij} = 2\mu e_{ij} + \delta_{ij} (\lambda e_{kk} - \gamma_t T - \gamma_n N - \gamma_m M), \quad (1)$$

where t_{ij} and e_{ij} are the stress and strain tensors, respectively; λ and μ are Lamé's constants; δ_{ij} is the Kronecker delta; T denotes the temperature change from a reference state; N is the variation in carrier density; and M is the moisture concentration. The coupling coefficients are defined as $\gamma_t = (3\lambda + 2\mu)\alpha_t$, $\gamma_n = (3\lambda + 2\mu)\alpha_n$, and $\gamma_m = (3\lambda + 2\mu)\alpha_m$, corresponding to thermal expansion, electronic deformation (carrier-thermal coupling), and moisture expansion, with α_t , α_n , and α_m being the respective material constants. The parameter $\xi = (e_0 a)^2$ represents the spatial nonlocal effect, where e_0 is a material-specific constant and a is an internal characteristic length. In the limit $\xi \rightarrow 0$, the model reduces to the classical local theory. The operator $(1 - \xi^2 \nabla^2 + \tau^2 \partial^2 / \partial t^2)$, which includes both spatial and temporal nonlocal terms, introduces scale dependency into the mechanical response and ensures finite propagation speeds for coupled disturbances.

The equation of motion for the displacement vector u_i , incorporating the same nonlocal operator, is given by:

$$2\mu e_{ij,j} + \lambda e_{kk,i} - \gamma_t T_{,i} - \gamma_n N_{,i} - \gamma_m M_{,i} = \rho \left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) \frac{\partial^2 u_i}{\partial t^2}, \quad (2)$$

where ρ is the mass density, a comma denotes a partial derivative, and a superposed dot indicates differentiation with respect to time.

The carrier density evolution is governed by a diffusion-recombination equation that includes thermal generation [8, 9]:

$$\frac{\partial N}{\partial t} = D_e N_{,ii} - \frac{N}{\tau_c} + \delta \frac{T}{\tau_c}, \quad (3)$$

with D_e the carrier diffusion coefficient, τ_c the carrier lifetime, $\delta = \partial n_0 / \partial T$ the thermal activation parameter, and n_0 the equilibrium carrier concentration.

The Modified of the MGT heat conduction equation for coupled hygro-photo-thermoelasticity [43]:

$$\left(D_t (\dot{T}_{,j})_{,i} + D_t^* (T_{,j})_{,i} \right) + D_t^m M_{,ij} + \frac{E_g \dot{N}}{\rho C_e \tau_c} = \left(1 + \tau_q \frac{\partial}{\partial t} \right) \left[\ddot{T} + \frac{\gamma_t T_o}{\rho C_e} \ddot{\epsilon}_{kk} \right], \quad (4)$$

where $D_t = K / (\rho C_e)$ is the thermal diffusivity, K the thermal conductivity, C_e the specific heat at constant strain, E_g the semiconductor energy gap, T_o the reference temperature, D_t^m the moisture-thermal coupling diffusivity, and τ_q the thermal relaxation time.

Similarly, Equation of moisture diffusion for coupled hygro-photo-thermoelasticity is given by [46]:

$$K_m \left(D_m (\dot{M}_{,j})_{,i} + D_m^* (M_{,j})_{,i} \right) + K_m D_m^t T_{,ij} + \frac{E_g \dot{N}}{\tau_c} = \left(1 + \tau_m \frac{\partial}{\partial t} \right) [K_m \ddot{M} + \gamma_m D_m M_o \ddot{\epsilon}_{kk}], \quad (5)$$

in which K_m is the moisture diffusion constant, D_m the moisture diffusivity, D_m^t the thermal-moisture coupling coefficient, M_o the reference moisture concentration, and τ_m the moisture relaxation time.

This study considers three primary formulations of fractional derivatives $D_t^{(\alpha)} := \frac{\partial^\alpha}{\partial t^\alpha}$. The standard Caputo time-fractional derivative is defined as [47, 48]:

$${}^C D_t^{(\alpha)} f(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t \frac{1}{(t-s)^\alpha} \frac{df(s)}{ds} ds, \quad 0 < \alpha < 1. \quad (6)$$

Following Caputo and Fabrizio [50,52], the fractional derivative of order α is expressed as:

$${}^{CF} D_t^{(\alpha)} f(t) = \frac{1}{1-\alpha} \int_0^t \exp\left(-\frac{\alpha(t-s)}{1-\alpha}\right) \frac{df(s)}{ds} ds, \quad 0 < \alpha < 1. \quad (7)$$

Recent literature suggests that the Caputo–Fabrizio (CF) derivative may function more as a filter than a conventional fractional derivative, as its localized kernel may not fully represent the complex long-range dynamics inherent in heat flow. In contrast, non-local frameworks employing the extended ML function are better suited for describing elasticity and non-local memory effects. The AB fractional time derivative [53] is given by:

$${}^{AB} D_t^{(\alpha)} f(t) = \frac{1}{1-\alpha} \int_0^t E_\alpha\left(-\frac{\alpha}{1-\alpha}(t-s)^\alpha\right) \frac{df(s)}{ds} ds, \quad 0 < \alpha < 1. \quad (8)$$

where the ML function $E_\alpha(z)$ is defined as:

$$E_\alpha(z) = \sum_{r=0}^{\infty} \frac{z^r}{\Gamma(1+r\alpha)}, \quad z \in \mathbb{C}, \quad 0 < \alpha < 1. \quad (9)$$

By replacing the ordinary time derivatives in Eqs. (4) and (5) with the AB fractional time derivative, the Fourier law is converted into an improved fractional-order formulation:

$$\left(D_t (\dot{T}_{,j})_{,i} + D_t^* (T_{,j})_{,i} \right) + D_t^m M_{,ij} + \frac{E_g \dot{N}}{\rho C_e \tau_c} = \left(1 + \tau_q {}^C D_t^{(\alpha)} \right) \left[\ddot{T} + \frac{\gamma_t T_o}{\rho C_e} \ddot{\epsilon}_{kk} \right], \quad (10)$$

$$K_m \left(D_m (\dot{M}_{,j})_{,i} + D_m^* (M_{,j})_{,i} \right) + K_m D_m^t T_{,ij} + \frac{E_g \dot{N}}{\tau_c} = \left(1 + \tau_m {}^C D_t^{(\alpha)} \right) [K_m \ddot{M} + \gamma_m D_m M_o \ddot{\epsilon}_{kk}]. \quad (11)$$

Equations (10) and (11) are advantageous for modeling heat propagation in heterogeneous media, biological tissues, polymers, porous and moisture-laden fractal structures, because they capture anomalous diffusion and non-local memory effects that classical diffusion frameworks cannot. Together with Eqs. (1)–(3), they form a coupled framework that integrates spatiotemporal nonlocality, elastic deformation, thermal relaxation, photogenerated carrier dynamics, and moisture diffusion. The nonlocal operator is central to

capturing microstructural effects, enabling accurate predictions of wave propagation, stress concentration, and energy transport at small scales where classical theories fall short, thus providing a robust model for complex material systems.

3. Formulation of the problem

The semiconductor medium is modeled as a hollow cylinder with inner radius a and outer radius b , assumed to be thermally homogeneous and isotropic. To characterize the photothermal response, the outer surface of the hollow cylinder, $r = b$, is irradiated by a laser pulse, while the inner surface, $r = a$, is subjected to a compressive step pressure. This loading configuration is retained consistently throughout the formulation and subsequent analysis. The analysis is performed in a cylindrical coordinate system (r, θ, z) , with the z -axis aligned with the cylinder axis. Initially, the medium is maintained at a uniform constant temperature T_0 . Within the MGT framework and subject to the regularity requirement, all physical fields are taken to remain finite. Owing to the geometric symmetry and the nature of the loading, the problem reduces to one dimension, so that all field variables depend only on the radial coordinate r and time t . As a result, the coupled governing equations, incorporating nonlocal elasticity, moisture diffusion, and MGT-based heat conduction, are formulated to encapsulate the radial evolution of displacement, thermal temperature, carrier density, and moisture concentration:

$$u = (u(r,t), 0, 0), \quad T = T(r,t), \quad N = N(r,t), \quad M = M(r,t). \quad (12)$$

Consequently, the strain components are

$$e_{rr} = \frac{\partial u}{\partial r}, \quad e_{\theta\theta} = \frac{u}{r}, \quad e_{zz} = e_{r\theta} = e_{rz} = e_{\theta z} = 0. \quad (13)$$

Furthermore, the cubic dilatation e and the Laplacian operator ∇^2 are defined as:

$$e = \frac{\partial u}{\partial r} + \frac{u}{r} = \frac{1}{r} \frac{\partial(ru)}{\partial r}, \quad \nabla^2 = \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r}. \quad (14)$$

By substituting the field representations into the nonlocal constitutive relation along with the corresponding governing equations for motion, carrier diffusion, heat conduction, and moisture transport, we establish the explicit field equations for the nonlocal hygro-photo-thermoelastic medium. This integration results in a cohesive system that effectively captures the coupled dynamics within the moisture-affected hollow cylinder.

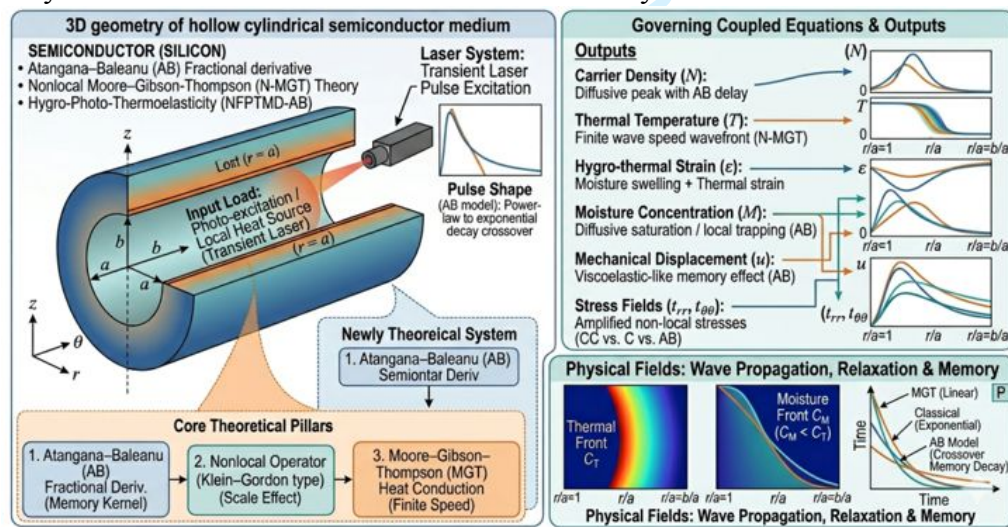


Figure 1: A schematic diagram of the three-dimensional geometry of the hollow cylindrical semiconductor medium is presented in the original work

By applying the MGT framework to these relations, the resulting system effectively incorporates relaxation effects alongside spatiotemporal nonlocality. This comprehensive approach leads to the formulation of the following governing equations:

$$\left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) t_{rr} = 2\mu \frac{\partial u}{\partial r} + \lambda e - \gamma_t T - \gamma_n N - \gamma_m M, \quad (15)$$

$$\left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) t_{\theta\theta} = 2\mu \frac{u}{r} + \lambda e - \gamma_t T - \gamma_n N - \gamma_m M, \quad (16)$$

$$\left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) t_{zz} = \lambda e - \gamma_t T - \gamma_n N - \gamma_m M, \quad (17)$$

$$(\lambda + 2\mu) \frac{\partial e}{\partial r} - \gamma_t \frac{\partial T}{\partial r} - \gamma_n \frac{\partial N}{\partial r} - \gamma_m \frac{\partial M}{\partial r} = \rho \left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) \frac{\partial^2 u}{\partial t^2}, \quad (18)$$

$$\frac{\partial N}{\partial t} = D_e \nabla^2 N - \frac{N}{\tau_c} + \delta \frac{T}{\tau_c}, \quad (19)$$

$$\begin{aligned} & K_m \left[D_m \frac{\partial}{\partial t} \nabla^2 M + D_m^* \nabla^2 M \right] + K_m D_m^t \nabla^2 T + \frac{E_g \dot{N}}{\tau_c} \\ &= (1 + \tau_m^\alpha D_t^{(\alpha)}) \left[K_m \frac{\partial^2 M}{\partial t^2} + \gamma_m D_m M_o \frac{\partial^2 e}{\partial t^2} \right], \end{aligned} \quad (20)$$

$$\begin{aligned} & \left[K \frac{\partial}{\partial t} \nabla^2 T + K^* \frac{\partial}{\partial t} \nabla^2 T + \rho C_e D_t^m \nabla^2 M \right] + \frac{E_g \dot{N}}{\tau_c} \\ &= (1 + \tau_q^\alpha D_t^{(\alpha)}) \left[\rho C_e \frac{\partial^2 T}{\partial t^2} + \gamma_t T_o \frac{\partial^2 e}{\partial t^2} \right] \end{aligned} \quad (21)$$

Applying the operator $\frac{1}{r} + \frac{\partial}{\partial r}$ on both sides of eq. (18), we obtain the following

$$(\lambda + 2\mu) \nabla^2 e - \gamma_t \nabla^2 T - \gamma_n \nabla^2 N - \gamma_m \nabla^2 M = \rho \left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) \frac{\partial^2 e}{\partial t^2}. \quad (22)$$

4. Solution of the problem

4.1 Dimensionless of Governing Equations

To facilitate the analysis and generalize the governing equations for the nonlocal hygro-photo-thermoelastic model, we introduce a set of dimensionless variables. The transformation of physical quantities into their dimensionless counterparts is defined as follows:

$$\begin{aligned} (r', u', \xi') &= \eta_1 C_o(r, u, \xi), \quad (t'_{rr}, t'_{\theta\theta}, t'_{zz}) = \frac{1}{\lambda + 2\mu} (t_{rr}, t_{\theta\theta}, t_{zz}), \quad N' = \frac{\gamma_n}{\lambda + 2\mu} N, \\ (t', \tau_m^\alpha, \tau_q^\alpha, \tau') &= \eta_1 C_o^2(t, \tau_m^\alpha, \tau_q^\alpha, \tau), \quad T' = \frac{\gamma_t}{\lambda + 2\mu} T, \quad M' = \frac{\gamma_m}{\lambda + 2\mu} M, \end{aligned} \quad (23)$$

$$\text{where } \eta_1 = \frac{\rho C_e}{K}, \quad C_o^2 = \frac{\lambda + 2\mu}{\rho}, \quad \gamma = \sqrt{\frac{\mu}{\lambda + 2\mu}}.$$

By substituting these dimensionless variables into the original governing equations (15)–(22) and omitting the primes for brevity, we derive the normalized field equations for the nonlocal hygro-photo-thermoelastic medium as follows:

$$\nabla^2 (e - T - N - M) = \left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) \frac{\partial^2 e}{\partial t^2}, \quad (24)$$

$$\frac{\partial N}{\partial t} = p_1 \nabla^2 N - p_2 N + p_3 T, \quad (25)$$

$$\nabla^2 \left(\frac{\partial M}{\partial t} + p_4 M + p_5 T \right) = -p_6 \frac{\partial N}{\partial t} + \left(1 + \tau_m^\alpha \frac{\partial^\alpha}{\partial t^\alpha}\right) \left[p_7 \frac{\partial^2 M}{\partial t^2} + p_8 \frac{\partial^2 e}{\partial t^2} \right], \quad (26)$$

$$\nabla^2 \left(\frac{\partial T}{\partial t} + p_9 T + p_{10} M \right) = -p_{11} \frac{\partial N}{\partial t} + \left(1 + \tau_q^\alpha \frac{\partial^\alpha}{\partial t^\alpha}\right) \left[p_{12} \frac{\partial^2 T}{\partial t^2} + p_{13} \frac{\partial^2 e}{\partial t^2} \right], \quad (27)$$

$$\left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) t_{rr} = 2\gamma^2 \frac{\partial u}{\partial r} + (1 - 2\gamma^2) e - (T + N + M), \quad (28)$$

$$\left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) t_{\theta\theta} = 2\gamma^2 \frac{u}{r} + (1 - 2\gamma^2) e - (T + N + M), \quad (29)$$

$$\left(1 - \xi^2 \nabla^2 + \tau^2 \frac{\partial^2}{\partial t^2}\right) t_{zz} = (1 - 2\gamma^2) e - (T + N + M), \quad (30)$$

For ease of reference, all dimensionless coefficients introduced in Eqs. (24)-(30) are consolidated in Table 1 below, together with their definitions and physical meanings. This table serves as a single reference for the remainder of the derivation.

Table 1. Consolidated summary of the dimensionless coefficients and their physical meanings.

Symbol	Definition	Physical meaning
p_1	$D_e \eta_1$	Ratio of carrier diffusion to characteristic transport rate
p_2	$1/(\eta_1 C_0^2 \tau_c)$	Inverse dimensionless carrier lifetime
p_3	$\delta \gamma_n/(\gamma_t \eta_1 C_0^2 \tau_c)$	Thermal-to-carrier activation coupling
p_4	$D_m^*/(D_m \eta_1 C_0^2)$	Ratio of moisture diffusivity rate to moisture diffusivity
p_5	$D_m^t \gamma_m/(D_m \eta_1 C_0^2 \gamma_t)$	Thermal–moisture cross-coupling coefficient
p_6	$\frac{E_g \gamma_m}{K_m D_m \tau_c \gamma_n \eta_1^2 C_0^2}$	Carrier–moisture coupling via recombination energy
p_7	$1/(D_m \eta_1)$	Inverse dimensionless moisture diffusivity
p_8	$\frac{\gamma_m^2 M_0}{K_m \eta_1 (\lambda + 2\mu)}$	Moisture–elastic coupling strength
p_9	K^*/K	Ratio of thermal conductivity rate to thermal conductivity
p_{10}	$\frac{\rho C_e D_m^t \gamma_t}{K \gamma_m \eta_1 C_0^2}$	Moisture–thermal coupling in the heat equation
p_{11}	$\frac{E_g \gamma_t}{K \tau_c \gamma_n \eta_1^2 C_0^2}$	Carrier–thermal coupling via recombination energy
p_{12}	$\rho C_e/(K \eta_1)$	Inverse dimensionless thermal diffusivity
p_{13}	$\frac{\gamma_t^2 T_0}{K \eta_1 (\lambda + 2\mu)}$	Thermoelastic coupling strength

The system is supplemented with homogeneous initial conditions, reflecting that the medium is initially at rest and in an unperturbed state:

$$u(r,0) = 0 = \frac{\partial u(r,0)}{\partial t}, T(r,0) = 0 = \frac{\partial T(r,0)}{\partial t}, \tag{31}$$

$$N(r,0) = 0 = \frac{\partial N(r,0)}{\partial t}, M(r,0) = 0 = \frac{\partial M(r,0)}{\partial t}. \tag{32}$$

4.2 Laplace transformation

For any function $\mathcal{F}(r,t)$, the Laplace transform is defined as:

$$\mathcal{L}(\mathcal{F}(r,t)) = \bar{\mathcal{F}}(r,s) = \int_0^\infty \mathcal{F}(r,t) e^{-st} dt, \tag{33}$$

where s is the complex frequency parameter.

By applying the Laplace transform defined in equation (33) to the dimensionless field equations (24)–(30), and incorporating the homogeneous initial conditions specified in (32), we derive the following transformed system of equations:

$$(p_{14} \nabla^2 - p_{15}) \bar{e} - \nabla^2 \bar{T} - \nabla^2 \bar{N} - \nabla^2 \bar{M} = 0, \tag{34}$$

$$(p_1 \nabla^2 - p_{16}) \bar{N} + p_3 \bar{T} = 0, \tag{35}$$

$$p_{17} \bar{e} + [p_{18} \nabla^2 - p_{19}] \bar{M} + p_5 \nabla^2 \bar{T} + s p_6 \bar{N} = 0, \tag{36}$$

$$p_{20} \bar{e} + [p_{21} \nabla^2 - p_{22}] \bar{T} + p_{10} \nabla^2 \bar{M} + s p_{11} \bar{N} = 0, \tag{37}$$

$$(1 - \xi^2 \nabla^2 + \tau^2 s^2) \bar{t}_{rr} = 2 \gamma^2 \frac{d\bar{u}}{dr} + (1 - 2 \gamma^2) \bar{e} - (\bar{T} + \bar{N} + \bar{M}), \tag{38}$$

$$(1 - \xi^2 \nabla^2 + \tau^2 s^2) \bar{t}_{\theta\theta} = 2\gamma^2 \frac{\bar{u}}{r} + (1 - 2\gamma^2) \bar{e} - (\bar{T} + \bar{N} + \bar{M}), \quad (39)$$

$$(1 - \xi^2 \nabla^2 + \tau^2 s^2) \bar{t}_{zz} = (1 - 2\gamma^2) \bar{e} - (\bar{T} + \bar{N} + \bar{M}), \quad (40)$$

where

$$\begin{aligned} p_{14} &= (1 + s^2 \xi^2), p_{15} = s^2 (1 + \tau^2 s^2), p_{16} = p_2 + s, p_{17} = -s^2 p_8 \left(1 + \tau_m^\alpha \frac{s^\alpha}{\alpha + s^\alpha (1 - \alpha)}\right), \\ p_{18} &= p_4 + s, p_{19} = s^2 p_7 \left(1 + \tau_m^\alpha \frac{s^\alpha}{\alpha + s^\alpha (1 - \alpha)}\right), p_{20} = -s^2 p_{13} \left(1 + \tau_q^\alpha \frac{s^\alpha}{\alpha + s^\alpha (1 - \alpha)}\right), \\ p_{21} &= p_9 + s, p_{22} = s^2 p_{12} \left(1 + \tau_q^\alpha \frac{s^\alpha}{\alpha + s^\alpha (1 - \alpha)}\right), \nabla^2 = \frac{d^2}{dr^2} + \frac{1}{r} \frac{d}{dr}. \end{aligned} \quad (41)$$

Through systematic elimination of variables and sequential algebraic manipulations of equations (34)–(37), the coupled system governing the dilatational field is condensed into a single eighth-order homogeneous differential equation:

$$(\kappa_1 \nabla^8 + \kappa_2 \nabla^6 + \kappa_3 \nabla^4 + \kappa_4 \nabla^2 + \kappa_5) (\bar{e}, \bar{T}, \bar{N}, \bar{M}) = 0. \quad (42)$$

In this equation, the coefficients are defined in terms of the following intermediate parameters:

$$\begin{aligned} \kappa_1 &= (p_1 p_5 p_{10} p_{14} - p_1 p_{14} p_{18} p_{21}), \kappa_2 = -p_1 p_5 p_{10} p_{15} - p_5 p_{10} p_{14} p_{16} \\ &+ p_1 p_{10} p_{17} + p_1 p_5 p_{20} - p_1 p_{18} p_{20} - p_1 p_{17} p_{21} + p_1 p_{15} p_{18} p_{21} \\ &+ p_{14} p_{16} p_{18} p_{21} + p_1 p_{14} p_{19} p_{21} + p_1 p_{14} p_{18} p_{22}, \kappa_3 = -s p_3 p_6 p_{10} p_{14} \\ &+ p_5 p_{10} p_{15} p_{16} - p_3 p_{10} p_{17} - p_{10} p_{16} p_{17} + s p_3 p_{11} p_{14} p_{18} - p_5 p_{16} p_{20} \\ &+ p_3 p_{18} p_{20} + p_{16} p_{18} p_{20} + p_1 p_{19} p_{20} + p_{16} p_{17} p_{21} - p_{15} p_{16} p_{18} p_{21} \\ &- p_1 p_{15} p_{19} p_{21} - p_{14} p_{16} p_{19} p_{21} + p_1 p_{17} p_{22} - p_1 p_{15} p_{18} p_{22} \\ &- p_{14} p_{16} p_{18} p_{22} - p_1 p_{14} p_{19} p_{22}, \kappa_4 = s p_3 p_6 p_{10} p_{15} + s p_3 p_{11} p_{17} \\ &- s p_3 p_{11} p_{15} p_{18} - s p_3 p_{11} p_{14} p_{19} - s p_3 p_6 p_{20} - p_3 p_{19} p_{20} - p_{16} p_{19} p_{20} \\ &+ p_{15} p_{16} p_{19} p_{21} - p_{16} p_{17} p_{22} + p_{15} p_{16} p_{18} p_{22} + p_1 p_{15} p_{19} p_{22} \\ &+ p_{14} p_{16} p_{19} p_{22}, \kappa_5 = s p_3 p_{11} p_{15} p_{19} - p_{15} p_{16} p_{19} p_{22}. \end{aligned} \quad (43)$$

Equation (42) can be conveniently factorized as:

$$(\nabla^2 - m_1^2)(\nabla^2 - m_2^2)(\nabla^2 - m_3^2)(\nabla^2 - m_4^2) = 0. \quad (44)$$

For a hollow cylinder of either finite or infinite extent, the solution for any physical field variable must respect the cylindrical geometry. The general solution is accordingly expressed in terms of modified Bessel functions:

$$(\bar{e}, \bar{T}, \bar{N}, \bar{M})(r, s) = \sum_{j=1}^4 (1, \alpha_j, \beta_j, \gamma_j) [A_j(s) I_0(m_j r) + B_j(s) K_0(m_j r)]. \quad (45)$$

In this expression, $I_0(\cdot)$ and $K_0(\cdot)$ denote the modified Bessel functions of the first and second kind of order zero, respectively. The quantities m_j (with $j = 1, 2, 3, 4$) are the positive roots of the characteristic equation:

$$\kappa_1 m^8 + \kappa_2 m^6 + \kappa_3 m^4 + \kappa_4 m^2 + \kappa_5 = 0. \quad (46)$$

This characteristic equation is fundamentally influenced by the nonlocal parameters ξ and τ , underscoring their essential role in modifying wave dispersion and attenuation behavior. The coupling coefficients α_j , β_j , and γ_j represent the relative amplitudes of the carrier density, moisture concentration, and temperature fields with respect to the dilatational potential for each modal contribution. These coefficients are determined from the algebraic relations (34)–(37). For each characteristic root m_j , the coupling coefficients are explicitly given by:

$$\alpha_j = \frac{m_i^4 p_1 p_{10} p_{14} + m_i^2 (-p_1 p_{10} p_{15} - p_{10} p_{14} p_{16} + p_1 p_{20}) + (p_{10} p_{15} p_{16} - p_{16} p_{20})}{m_i^4 (p_1 p_{10} - p_1 p_{21}) + m_i^2 (-p_3 p_{10} - p_{10} p_{16} + p_{16} p_{21} + p_1 p_{22}) + (s p_3 p_{11} - p_{16} p_{22})}, \quad (47)$$

$$\beta_j = \frac{p_3 \alpha_i}{-m_i^2 p_1 + p_{16}}, \quad (48)$$

$$\gamma_j = \frac{m_i^2 p_{14} - m_i^2 (\alpha_i + \beta_i) - p_{15}}{m_i^2}. \quad (49)$$

In the Laplace-transformed domain, the radial displacement $\bar{u}$ can be expressed as:

$$\bar{u} = \sum_{j=1}^4 \frac{1}{m_j} [A_j(s) I_1(m_j r) - B_j(s) K_1(m_j r)], \quad (50)$$

where $I_1(\cdot)$ and $K_1(\cdot)$ are the modified Bessel functions of the first and second kind of order one. Equation (50) is derived using the standard Bessel function integration relations:

$$\int r I_0(r) dr = r I_1(r), \text{ \& } \int r K_0(r) dr = -r K_1(r). \quad (51)$$

Differentiating equation (50) with respect to the radial coordinate r yields:

$$\frac{d\bar{u}}{dr} = \sum_{j=1}^4 A_j(s) \left[I_0(m_j r) - \frac{1}{m_j r} I_1(m_j r) \right] + B_j(s) \left[K_0(m_j r) + \frac{1}{m_j r} K_1(m_j r) \right]. \quad (52)$$

Finally, by substituting the relevant expressions into equations (38)–(40), the stress components are obtained in terms of the modal amplitudes as follows:

$$\begin{aligned} \bar{t}_{rr} = & \sum_{j=1}^4 A_j(s) \left[\mathcal{H}_j I_0(m_j r) - \frac{2\gamma^2}{m_j r (1 - \xi^2 m_j^2 + \tau^2 s^2)} I_1(m_j r) \right] \\ & + \sum_{j=1}^4 B_j(s) \left[\mathcal{H}_j K_0(m_j r) + \frac{2\gamma^2}{m_j r (1 - \xi^2 m_j^2 + \tau^2 s^2)} K_1(m_j r) \right], \end{aligned} \quad (53)$$

$$\begin{aligned} \bar{t}_{\theta\theta} = & \sum_{j=1}^4 A_j(s) \left[\mathcal{F}_j I_0(m_j r) + \frac{2\gamma^2}{m_j r (1 - \xi^2 m_j^2 + \tau^2 s^2)} I_1(m_j r) \right] \\ & + \sum_{j=1}^4 B_j(s) \left[\mathcal{F}_j K_0(m_j r) - \frac{2\gamma^2}{m_j r (1 - \xi^2 m_j^2 + \tau^2 s^2)} K_1(m_j r) \right], \end{aligned} \quad (54)$$

$$\bar{t}_{zz} = \sum_{j=1}^4 [A_j(s) [\mathcal{F}_j I_0(m_j r)] + B_j(s) [\mathcal{F}_j K_0(m_j r)]], \quad (55)$$

where

$$\mathcal{H}_j = \frac{1 - \alpha_i - \beta_i - \gamma_i}{1 - \xi^2 m_j^2 + \tau^2 s^2}, \quad \mathcal{F}_j = \frac{1 - 2\gamma^2 - \alpha_i - \beta_i - \gamma_i}{1 - \xi^2 m_j^2 + \tau^2 s^2}.$$

4.3 Boundary conditions

To establish a complete mathematical model for the moisture-affected semiconductor housed within a finite hollow cylinder, it is imperative to define boundary conditions at both the inner surface ($r = a$) and the outer surface ($r = b$).

4.3.1. Mechanical Boundary Conditions

The internal boundary experiences a sudden compressive load, modeled as a step pressure, described mathematically by:

$$t_{rr}(a, t) = -p_i(t), \quad p_i(t) = p_0 H(t), \quad (56)$$

where $p_0 > 0$ represents the pressure magnitude and $H(t)$ is the Heaviside step function, defined as $H(t) = 0$ for $t < 0$ and $H(t) = 1$ for $t \geq 0$. The negative sign indicates that compressive radial stress acts inward at $r = a$, adhering to standard continuum mechanics conventions. The outer boundary is rigidly constrained in the radial direction, expressed as:

$$u(b, t) = 0. \quad (57)$$

4.3.2. Thermal Boundary Conditions

Thermal boundary conditions may prescribe either temperature or heat flux at the boundaries, depending on the thermal loading configuration. These are given by:

$$\left. \frac{\partial T}{\partial r} \right|_{r=a} = 0, \quad (58)$$

and

$$-k \left. \frac{\partial T}{\partial r} \right|_{r=b} = q_{laser}(t), \quad q_{laser}(t) = q_0 e^{-\frac{t}{\tau_p}}. \quad (59)$$

Here, $q_0 > 0$ denotes the peak heat flux, and $\tau_p > 0$ signifies the pulse decay time. The negative sign is in accordance with Fourier's law, ensuring that positive q_{laser} corresponds to heat flowing into the cylinder. While an exponentially decaying profile is used in this formulation, alternative temporal shapes such as Gaussian, rectangular, or sinusoidal pulses can be integrated without compromising the analytical framework.

4.3.3. Moisture Boundary Conditions

Both boundaries are assumed to be impermeable to moisture diffusion, resulting in zero-flux conditions expressed as:

$$\left. \frac{\partial M}{\partial r} \right|_{r=a} = 0, \quad \left. \frac{\partial M}{\partial r} \right|_{r=b} = 0. \quad (60)$$

4.3.4. Carrier-Density Boundary Conditions

Two physically representative configurations for carrier density are considered, with a specific choice dependent on the device environment. In this formulation, we adopt a no-flux condition at the inner surface and a surface recombination condition at the outer boundary, expressed as:

$$\left. \frac{\partial N}{\partial r} \right|_{r=a} = 0, \quad (61)$$

and

$$-D_e \left. \frac{\partial N}{\partial r} \right|_{r=b} = S_v N(b, t). \quad (62)$$

In this equation, S_v represents the surface recombination velocity (m/s), accounting for finite carrier recombination kinetics at $r = b$.

The boundary conditions defined in equations (56)–(62) are transformed into the Laplace s -domain using the transformation specified in equation (33), resulting in the following dimensionless relations:

At the inner surface ($r = a$):

$$\bar{t}_{rr} = -\frac{p_0}{s} = f_1(s), \quad \frac{d\bar{T}}{dr} = 0, \quad \frac{d\bar{M}}{dr} = 0, \quad \frac{d\bar{N}}{dr} = 0, \quad (63)$$

At the outer surface ($r = b$):

$$\bar{u} = 0, \quad \frac{d\bar{T}}{dr} = -\frac{q_0 \tau_p}{1+s\tau_p} = f_2(s), \quad \frac{d\bar{M}}{dr} = 0, \quad -D_e \frac{d\bar{N}}{dr} = s_v \bar{N}. \quad (64)$$

Substituting the general modal solutions into the boundary conditions (63) and (64) results in a closed system of eight linear algebraic equations for the unknown integration constants A_j (s) and $B_j(s)$ (where $j = 1, 2, 3, 4$):

$$\sum_{j=1}^4 [A_j(s) \mathcal{A}_{1j} + B_j(s) \mathcal{B}_{1j}] = f_1(s), \quad (65)$$

$$\sum_{j=1}^4 [A_j(s) \mathcal{A}_{2j} - B_j(s) \mathcal{B}_{2j}] = 0, \quad (66)$$

$$\sum_{j=1}^4 [A_j(s) \mathcal{A}_{3j} - B_j(s) \mathcal{B}_{3j}] = 0, \quad (67)$$

$$\sum_{j=1}^4 [A_j(s) \mathcal{A}_{4j} - B_j(s) \mathcal{B}_{4j}] = 0, \quad (68)$$

$$\sum_{j=1}^4 [A_j(s) \mathcal{A}_{5j} - B_j(s) \mathcal{B}_{5j}] = 0, \quad (69)$$

$$\sum_{j=1}^4 [A_j(s) \mathcal{A}_{6j} - B_j(s) \mathcal{B}_{6j}] = f_2(s), \quad (70)$$

$$\sum_{j=1}^4 [A_j(s) \mathcal{A}_{7j} - B_j(s) \mathcal{B}_{7j}] = 0, \quad (71)$$

$$\sum_{j=1}^4 [A_j(s) \mathcal{A}_{8j} - B_j(s) \mathcal{B}_{8j}] = 0. \quad (72)$$

In above equations, the coefficient matrices are explicitly defined as:

$$\begin{aligned} f_1(s) &= -\frac{p_0}{s}, \quad f_2(s) = -\frac{q_0 t_p}{1+s t_p}, \quad \mathcal{A}_{1j} = \mathcal{H}_j I_0(m_j a) - \frac{2\gamma^2}{m_j a (1-\xi^2 m_j^2 + \tau^2 s^2)} I_1(m_j a), \\ \mathcal{B}_{1j} &= \mathcal{H}_j K_0(m_j a) + \frac{2\gamma^2}{m_j a (1-\xi^2 m_j^2 + \tau^2 s^2)} K_1(m_j a), \quad \mathcal{A}_{2j} = \alpha_j m_j I_1(m_j a), \\ \mathcal{A}_{3j} &= \gamma_j m_j I_1(m_j a), \quad \mathcal{B}_{3j} = \gamma_j m_j K_1(m_j a), \quad \mathcal{A}_{4j} = \beta_j m_j I_1(m_j a), \quad \mathcal{B}_{4j} = \beta_j m_j K_1(m_j a), \\ \mathcal{A}_{5j} &= \frac{1}{m_j} I_1(m_j b), \quad \mathcal{B}_{5j} = \frac{1}{m_j} K_1(m_j b), \quad \mathcal{A}_{6j} = \alpha_j m_j I_1(m_j b), \quad \mathcal{B}_{6j} = \alpha_j m_j K_1(m_j b), \\ \mathcal{A}_{7j} &= \gamma_j m_j I_1(m_j b), \quad \mathcal{B}_{7j} = \gamma_j m_j K_1(m_j b), \quad \mathcal{A}_{8j} = D_e \beta_j m_j I_1(m_j b) + s_v \beta_j I_0(m_j b), \\ \mathcal{B}_{2j} &= \alpha_j m_j K_1(m_j a), \quad \mathcal{B}_{8j} = D_e \beta_j m_j K_1(m_j b) - s_v \beta_j K_0(m_j b). \end{aligned} \quad (73)$$

The system of equations (65)-(72) can be elegantly represented in matrix form as:

$$\mathbf{M}(s) \mathbf{X}(s) = \mathbf{F}(s), \quad (74)$$

where $\mathbf{X}(s) = [A_1, B_1, A_2, B_2, A_3, B_3, A_4, B_4]^T$ encompasses the unknown modal amplitudes. By employing standard techniques for matrix inversion, we can derive explicit expressions for $A_j(s)$ and $B_j(s)$ in the complex frequency domain. Once these constants are determined, they are substituted back into the general solutions, allowing us to reconstruct the complete Laplace-domain distributions of displacement, stress, temperature, carrier density, and moisture concentration throughout the cylinder wall.

To retrieve the transient physical fields in the time domain, a robust numerical inversion algorithm is necessary. Following the methodology established by Honig and Hirdes [38], we perform the inverse Laplace transform numerically with high precision. This crucial final step enables us to visualize the spatiotemporal evolution of the coupled fields, capturing wave reflections at the bounded surfaces. Additionally, it quantifies how the MGT relaxation mechanisms and spatiotemporal nonlocality influence finite-speed wavefront propagation within the semiconductor medium. This comprehensive approach not only elucidates the dynamic response but also enhances the understanding of multiphysical interactions at play.

5. Special cases

The proposed spatiotemporal nonlocal elasticity model, incorporating fractional time derivatives and coupled photo-induced carrier generation, thermal, moisture, and mechanical fields, provides a flexible framework for thermoelastic analysis in complex materials. It captures nonlocal interactions, thermal inertia, and memory effects, making it suitable for micro- and nanoscale systems where classical continuum theories are inadequate. A key advantage is its ability to recover established fractional and non-fractional thermoelastic models under specific parameter limits, thereby unifying and extending existing formulations. The resulting special cases are summarized in Table 2, each corresponding to particular parameter choices that isolate specific physical mechanisms or simplify the governing equations. This systematic reduction demonstrates the robustness of the proposed framework, confirms its consistency with established thermoelastic theories, and enhances predictive capability for moisture-sensitive, microstructured semiconductor systems.

Table 2. Summary of specialized models derived from the generalized hygro-photo-thermoelastic framework.

	Model description	Abbreviated name	Conditions
1	Full coupling: photo-thermal-hygro-mechanical diffusion (Classical Case)	PTHMD-CC	$\tau_q \neq 0, \tau_m \neq 0, \gamma_m \neq 0, \gamma_n \neq 0, \alpha = 1, \xi = 0, \tau = 0$
2	Photo-thermal-mechanical diffusion, no moisture (Classical Case)	PTMD-CC	$\tau_q \neq 0, \tau_m \neq 0, \gamma_m = 0, \gamma_n \neq 0, \alpha = 1, \xi = 0, \tau = 0$
3	Pure thermoelastic diffusion (Classical Case)	TMD-CC	$\tau_q \neq 0, \tau_m \neq 0, \gamma_m = 0, \gamma_n = 0, \alpha = 1, \xi = 0, \tau = 0$
4	Full coupling with spatiotemporal nonlocality	NPTHMD-CC	$\tau_q \neq 0, \tau_m \neq 0, \gamma_m \neq 0, \gamma_n \neq 0, \alpha = 1, \xi \neq 0, \tau \neq 0$
5	No-moisture model with spatiotemporal nonlocality	NPTMD-CC	$\tau_q \neq 0, \tau_m \neq 0, \gamma_m = 0, \gamma_n \neq 0, \alpha = 1, \xi \neq 0, \tau \neq 0$
6	Pure thermoelastic model with spatiotemporal nonlocality	NTMD-CC	$\tau_q \neq 0, \tau_m \neq 0, \gamma_m = 0, \gamma_n = 0, \alpha = 1, \xi \neq 0, \tau \neq 0$
7	Full coupling with spatiotemporal nonlocality and fractional derivative	NFPTHMD-C / NFPTHMD-AB	$\tau_q \neq 0, \tau_m \neq 0, \gamma_m \neq 0, \gamma_n \neq 0, \alpha < 1, \xi \neq 0, \tau \neq 0$
8	No-moisture model with spatiotemporal nonlocality and fractional derivative	NFPTMD-C / NFPTMD-AB	$\tau_q \neq 0, \tau_m \neq 0, \gamma_m = 0, \gamma_n \neq 0, \alpha < 1, \xi \neq 0, \tau \neq 0$
9	Pure thermoelastic model with spatiotemporal nonlocality and fractional derivative	NFTMD-C / NFTMD-AB	$\tau_q \neq 0, \tau_m \neq 0, \gamma_m = 0, \gamma_n = 0, \alpha < 1, \xi \neq 0, \tau \neq 0$

The suffixes -CC, -C, and -AB denote the Classical Case (integer-order derivative, $\alpha = 1$), the Caputo fractional derivative, and the Atangana–Baleanu fractional derivative, respectively. For the fractional cases (7–9), the same model abbreviation is retained for both operators, distinguished by the suffix -C or -AB, since the governing equations differ only in the choice of the time-fractional operator.

6. Numerical outcomes and interpretation

All numerical simulations are performed using material constants for crystalline silicon (Si), as summarized in Table 3. The reference time is fixed at $t = 0.005$, and the computational domain spans the radial interval $a \leq r \leq b$ corresponding to the hollow cylinder geometry.

Table 3. Material properties of silicon (Si) used in numerical simulations [31, 46].

Symbol	Quantity	Value	Symbol	Quantity	Value
λ	Lamé constant	$6.4 \times 10^{10} \text{ N/m}^2$	E_g	Energy gap	1.11 eV
μ	Shear modulus	$6.5 \times 10^{10} \text{ N/m}^2$	D_t^m	Moisture-thermal coupling diffusivity	2.1×10^{-7}
α_t	Thermal expansion coefficient	$4.14 \times 10^{-6} \text{ K}^{-1}$	T_0	Reference temperature	800 K
α_n	Electronic deformation coefficient	$-9 \times 10^{-31} \text{ m}^3$	K_m	Moisture diffusion constant	2.2×10^{-8}
α_m	Moisture expansion coefficient	$2.68 \times 10^{-3} \text{ cm}/(\text{cm} \cdot \% \text{H})$	D_m	Moisture diffusivity	$0.35 \times 10^{-11} \text{ m}^2/\text{s}$
ρ	Mass density	2330 kg/m^3	D_m^t	Thermal-moisture coupling coefficient	0.648×10^{-6}

D_e	Carrier diffusion coefficient	$2.5 \times 10^{-3} \text{ m}^2/\text{s}$	k	Thermal conductivity	$150 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$
τ_c	Carrier lifetime	$5 \times 10^{-5} \text{ s}$	n_0	Equilibrium carrier concentration	10^{10} m^{-3}
δ	Thermal activation parameter	$0.5 \text{ m}^{-3}\cdot\text{K}^{-1}$	M_0	Reference moisture concentration	10%
C_e	Specific heat at constant strain	$695 \text{ J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$	τ_q	Thermal relaxation time	0.01 s
—	—	—	τ_m	Moisture relaxation time	0.04 s

The spatial nonlocal parameter is defined as $\xi = (e_0 a)^2$, in which a represents the internal characteristic length and e_0 is a dimensionless calibration constant. For crystalline silicon, a corresponds to the lattice constant, approximately 0.543 nm. The constant e_0 is determined by matching the nonlocal dispersion relation to experimental phonon dispersion data or molecular-dynamics simulations. For semiconductors and nanostructured materials, the literature consistently reports e_0 in the range 0.5–2, which yields dimensional ξ values between about 0.07 nm² and 1.18 nm². Following the non-dimensionalization of Eq. (23), where ξ is scaled by $\eta_1 C_0$, these values correspond to the dimensionless range 0.01–0.10 adopted in the present study. The specific choices $\xi = 0.04$ and $\xi = 0.05$ lie near the middle of this admissible range and are representative of a moderately nonlocal semiconductor at micro- and nanoscale dimensions.

To assess parameter sensitivity, three comparative studies are performed. The first varies the spatiotemporal nonlocal parameters (ξ, τ_T), which represent microstructural length scales and inertial effects, to quantify changes in stress concentration, wave arrival, and peak amplitudes near the irradiated surface. The second examines laser pulse duration (t_p), showing how energy-deposition time controls the amplitude, penetration, and evolution of the coupled fields; shorter pulses produce sharper localized transients, whereas longer pulses yield smoother diffusive responses within the hyperbolic MGT framework. The third compares classical integer-order ($\alpha = 1$), Caputo fractional ($0 < \alpha < 1$), and AB fractional ($0 < \alpha < 1$) operators to assess how memory-dependent, non-singular kernels affect sub-diffusive transport, relaxation, and field coupling.

6.1. The effect of the nonlocal parameters on the physical fields

The following analysis investigates the influence of the spatiotemporal nonlocal parameters, specifically the spatial parameter ξ and the temporal relaxation time τ_T , on the distribution of physical fields within the hollow cylindrical semiconductor. By employing the nonlocal MGT theory using only the AB fractional derivative model (NFPTHMD-AB) with $\alpha = 0.8$, the analysis examines how these scaling factors modulate the coupled hygro-photo-thermoelastic response. The integration of these parameters allows the model to capture the finite speed of wave propagation and the interatomic long-range forces, which are essential for the structural assessment of micro-scale cylinders under transient laser-pulse loading.

Figure 2 illustrates the effect of nonlocal parameters on the temperature T across the radial distance r . All curves initiate at a near-zero state at the internal boundary ($r = r_1$) and exhibit a monotonic increase toward the outer boundary. Quantitatively, the temperature reaches its maximum values as r approaches r_2 . Physically, this behavior represents the thermal wave propagation through the semiconductor medium in response to internal photo-excitation. The MGT framework ensures that heat propagates at a finite speed, preventing the instantaneous thermal jumps associated with classical Fourier heat conduction. In practice, this allows for

precise thermal management in MEMS devices, ensuring that moisture-sensitive semiconductor layers remain within safe operating temperatures.

Figure 3 presents the effect of nonlocal parameters on the radial displacement u , which remains negative throughout the radial range, signifying an inward or compressive deformation. The magnitude of displacement drops sharply near the inner boundary before recovering toward zero. Quantitatively, the spatiotemporal parameters exert a dominant effect: the combination of $\xi \neq 0$ and $\tau_T \neq 0$ produces the most significant displacement, whereas the local-like model with $\xi = 0$ shows the smallest magnitude. Physically, this displacement is the mechanical consequence of the combined plasma pressure and thermal expansion. The nonlocal parameter acts as a spatial scaling agent that incorporates interatomic interactions, which in this case amplifies the total deformation.

Figure 4 displays the effect of nonlocal parameters on the carrier density N as it propagates from the inner surface. The distribution is characterized by an initial fluctuation near $r = r_1$, followed by a peak in the intermediate region. Quantitatively, the classical local model ($\xi = 0$, $\tau_T = 0$) starts with the highest initial carrier density at the boundary, while the nonlocal models exhibit a lower initial state but show more distinct peaks, particularly when the fractional relaxation is applied. Physically, this represents the diffusion and recombination of photo-excited electrons and holes. The nonlocal fractional MGT theory introduces a memory effect in the carrier transport, causing a shift and a sharpening of the density peak compared to local theory, which is essential for predicting the electrical response time of the device.

Figure 5 examines the effect of nonlocal parameters on the moisture concentration M , which rises steadily as the radial distance increases. A clear hierarchical separation between the curves is observed based on the nonlocal parameters. Quantitatively, the case where $\xi \neq 0$ and $\tau_T \neq 0$ yields the highest moisture levels, whereas the absence of the relaxation parameter ($\tau_T = 0$) yields the lowest. Physically, this illustrates the hygro aspect of the model, where moisture diffuses through the porous semiconductor matrix. The nonlocal parameters account for the fact that moisture transport at the micro-scale depends on the surrounding structural environment, a vital factor for the reliability of environmental sensors operating in high-humidity conditions.

Figure 6 depicts the effect of nonlocal parameters on the radial stress t_{rr} . The stress t_{rr} starts at zero at the internal surface, satisfying the boundary conditions, and becomes increasingly compressive (negative) as it moves outward. Quantitatively, the curve with higher spatiotemporal nonlocality ($\xi \neq 0$, $\tau_T \neq 0$) produces the maximum compressive stress, while the curve without spatial nonlocality maintains the lowest stress magnitude. Physically, σ_{rr} results from the internal pressure generated by moisture concentration and thermal gradients. The results demonstrate that neglecting nonlocality leads to an underestimation of the compressive stress within the cylinder, which is essential for preventing structural failure in high-pressure cylindrical components.

Figure 7 shows the effect of nonlocal parameters on the hoop stress $t_{\theta\theta}$, which starts with high negative values at the inner boundary. Most curves show a characteristic recovery or stabilization after the initial drop. Quantitatively, varying the spatiotemporal parameters significantly alters the initial stress intensity at the laser-impact site. Physically, the hoop stress is often the primary cause of longitudinal cracks in cylindrical geometries. The nonlocal MGT theory provides a more realistic representation of the stress concentration by distributing the load across the interatomic lattice, providing engineers the necessary data to optimize the wall thickness of hollow cylinders to withstand circumferential compression.

Before proceeding, two physically distinct aspects of spatial nonlocality must be separated, since conflating them can lead to apparent contradictions in the interpretation of the stress fields. The nonlocal operator smooths the sharpest local gradients by distributing the mechanical response over the internal characteristic length. This broadens the region over

which deformation is transmitted and eliminates the unphysical gradient singularities that local theories may predict near a loaded surface. Such redistribution, however, does not necessarily reduce the magnitudes of the individual stress components. For the present hollow cylinder, which combines a compressive step load at the inner surface with a rigidly constrained outer surface, the numerical results in Figures 6 and 7 show that increasing the spatiotemporal nonlocal parameters ξ and τ_T consistently increases the magnitudes of both the compressive radial stress and the compressive hoop stress. Consequently, as explicitly stated in Section 7, local theories that neglect spatial nonlocality underestimate the compressive stress state of the cylinder. This behavior is configuration-dependent, reflecting the interplay between nonlocal strain redistribution and the kinematic constraints imposed at the two cylindrical surfaces, and it should not be generalized to geometries or boundary conditions for which the nonlocal redistribution may act in the opposite sense.

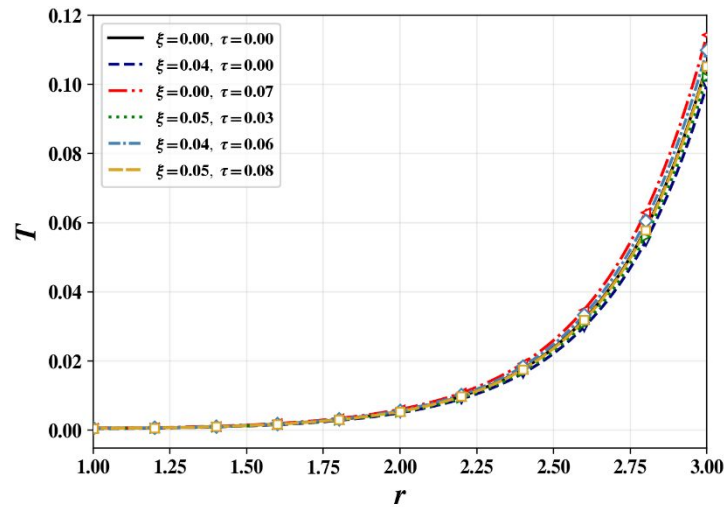


Figure 2: Effect of nonlocal parameters on the temperature (T).

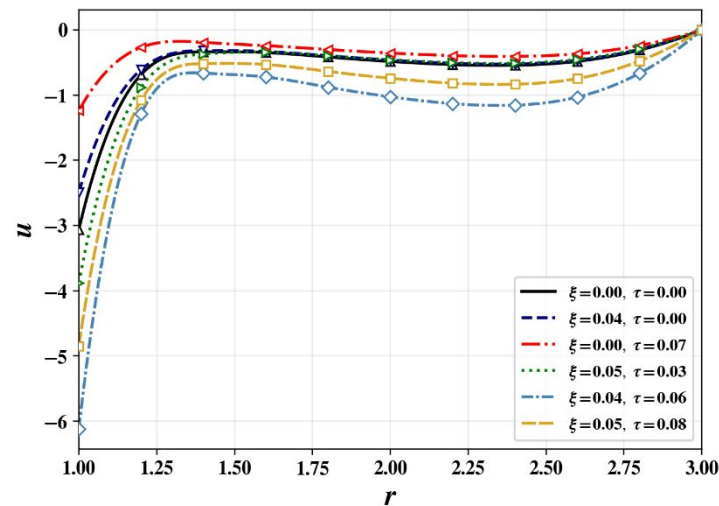


Figure 3: Effect of nonlocal parameters on the radial displacement (u).

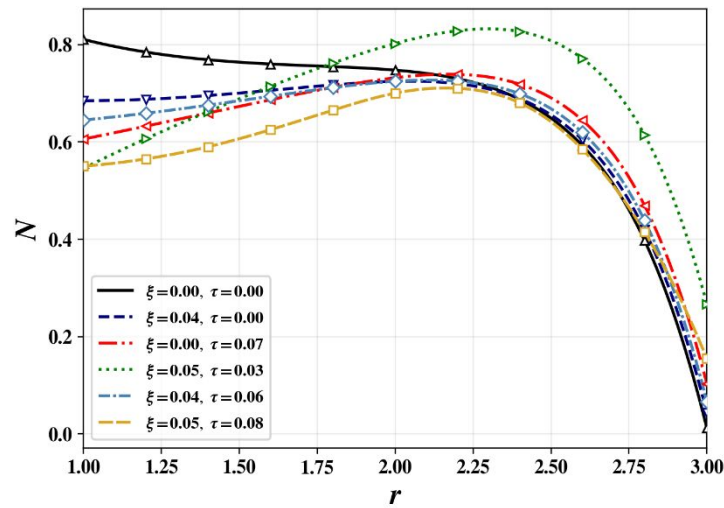


Figure 4: Effect of nonlocal parameters on the carrier density (N).

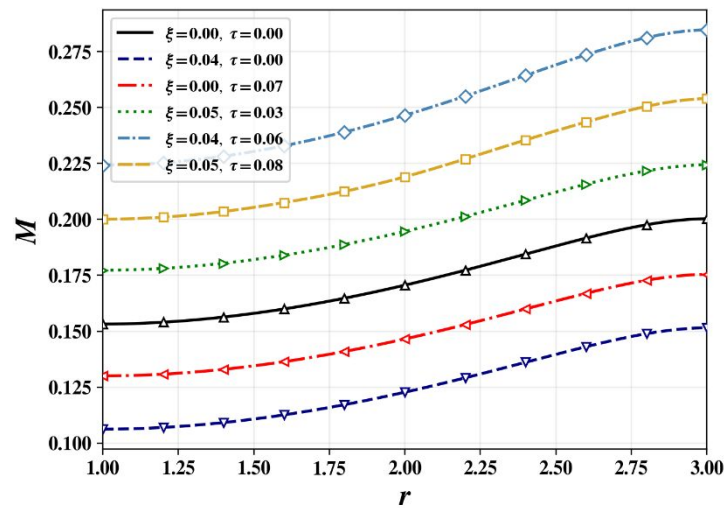


Figure 5: Effect of nonlocal parameters on the moisture (M).

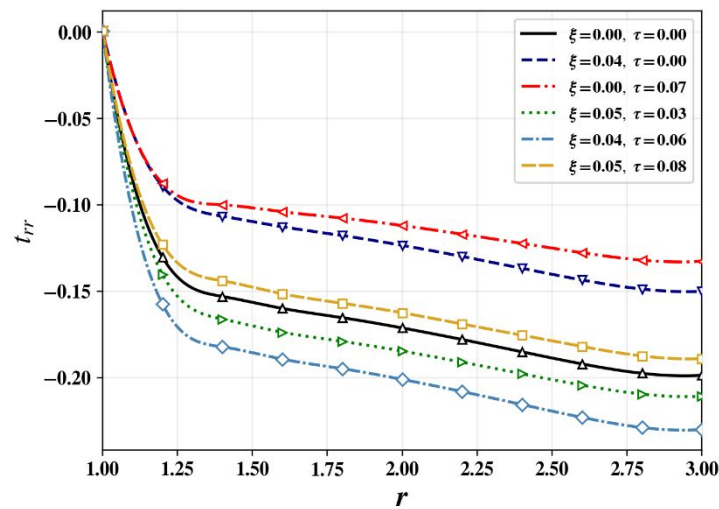


Figure 6: Effect of nonlocal parameters on the radial stress (t_{rr}).

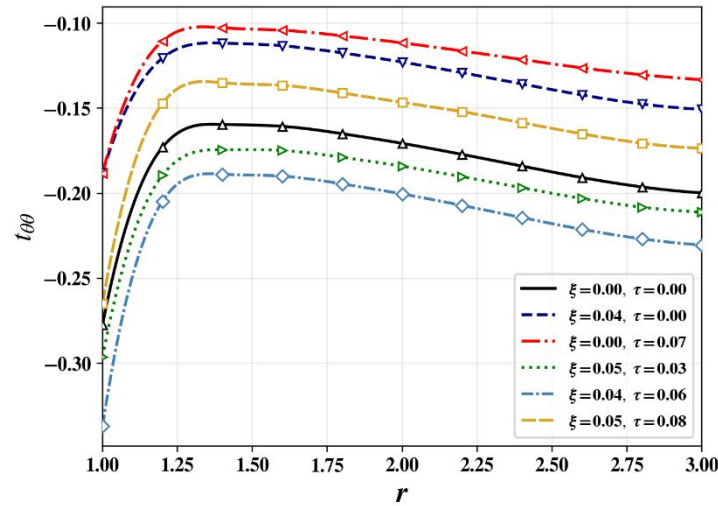


Figure 7: Effect of nonlocal parameters on the hoop stress ($t_{\theta\theta}$).

6.2. Influence laser pulses on Physical Fields

The dynamic response of a moisture-affected, photo-thermoelastic semiconductor medium to transient laser heating is profoundly sensitive to the temporal profile of the laser pulse, particularly its duration t_p . Within the framework of the nonlocal fractional MGT theory, this section specifically employs the AB fractional derivative model (NFPTHMD-AB) with $\alpha = 0.8$ to investigate these coupled interactions. By utilizing the Laplace transform to handle the time-dependent governing equations, the model captures the nonlocal and non-singular kernel characteristics inherent in the AB derivative, providing a more precise representation of the memory effects and thermal delays compared to classical fractional operators. Five pulse durations are considered, $t_p = 0.1, 0.2, 0.3, 0.4$ and 0.5 , spanning the transition from a strongly transient to a quasi-steady thermal load. Throughout the following discussion, $t_p = 0.1$ corresponds to the shortest and most intense pulse, while $t_p = 0.5$ corresponds to the longest and most diffuse. Figures 8 through 13 provide a comprehensive illustration of how varying t_p modulates the spatial distributions of displacement, stress, temperature, carrier density, and moisture concentration within the hollow cylindrical medium.

Figure 8 shows the effect of pulse duration on the temperature T along the radial coordinate r . All curves begin near zero at the inner surface and rise nonlinearly toward the irradiated outer boundary, with the steepest growth confined to $r > 2.2$. Quantitatively, the curves are practically indistinguishable for $r < 1.8$, and separate progressively thereafter: at $r = 3$ the temperature falls monotonically from $T \approx 0.131$ for $t_p = 0.1$ to $T \approx 0.096$ for $t_p = 0.5$, a reduction of roughly 27%. Physically, a shorter pulse concentrates the deposited energy over a briefer interval, so the thermal disturbance retains a sharper front and a higher peak; longer pulses allow the MGT relaxation mechanism more time to redistribute the energy, flattening the profile. In practice this indicates that stretching the pulse is an effective means of limiting localized thermal loading near the irradiated face of a photonic device.

Figure 9 presents the radial displacement u , which is negative throughout, denoting inward (compressive) deformation. Each curve attains its largest magnitude at the inner surface, recovers steeply to a local minimum magnitude near $r \approx 1.45$, passes through a secondary trough around $r \approx 2.35$, and vanishes at $r = 3$ in accordance with the clamped condition $u(b, t) = 0$. Quantitatively, the inner-surface magnitude decreases monotonically from $|u| \approx 7.2$ at $t_p = 0.1$ to $|u| \approx 5.5$ at $t_p = 0.5$, a reduction of about 24%, while the secondary trough

diminishes from ≈ 1.35 to ≈ 1.1 . Physically, the deformation reflects the combined plasma pressure and thermal expansion; shorter pulses deposit energy faster than the lattice can relax, producing larger transient strain. For MEMS design, this establishes pulse width as a direct control parameter for peak actuator deflection.

Figure 10 tracks the carrier density N . Each distribution rises gently from the inner surface to a broad maximum near $r \approx 2.1$ and then decreases sharply toward the outer boundary. Quantitatively, the peak value drops from $N \approx 0.87$ at $t_p = 0.1$ to $N \approx 0.61$ at $t_p = 0.5$, about 30%, while all five curves converge to $N \approx 0.06$ at $r = 3$, a direct consequence of the surface-recombination condition (61). Physically, shorter pulses drive photogeneration faster than diffusion and bulk recombination can deplete the excess carriers, sustaining a higher plasma density; longer pulses give electron-hole pairs time to recombine as they are generated. The convergence at $r = b$ shows that the recombination velocity S_v , not the pulse profile, fixes the boundary carrier level.

Figure 11 shows the moisture concentration M , which increases smoothly with r for every pulse duration. The curves are vertically ordered without crossing: M ranges from 0.27 to 0.34 for $t_p = 0.1$ and from 0.193 to 0.245 for $t_p = 0.5$, an overall downward shift of roughly 28%. Physically, the hygral field is slaved to the thermal and plasma fields through the cross-coupling terms in Eq. (20); the more intense transients generated by short pulses enhance moisture redistribution and local accumulation within the semiconductor matrix. For devices operating in humid environments, this suggests that short-pulse regimes elevate moisture-induced swelling and the associated internal pressure.

Figure 12 displays the radial stress t_{rr} , which vanishes at the inner surface, falls steeply through a knee at $r \approx 1.3$, and then declines gradually toward the outer boundary, remaining compressive throughout. Quantitatively, the outer-boundary value decreases in magnitude monotonically with pulse duration, from $t_{rr} \approx -0.277$ at $t_p = 0.1$ to $t_{rr} \approx -0.199$ at $t_p = 0.5$, a reduction of about 28%. Physically, t_{rr} is the mechanical consequence of the volumetric expansion driven jointly by temperature, plasma density, and moisture; rapid deposition magnifies the internal compressive load because relaxation cannot keep pace. Lengthening the pulse fivefold therefore relieves roughly a quarter of the peak compressive load.

Figure 13 presents the hoop stress $t_{\theta\theta}$, which is strongly compressive at the inner boundary, relaxes to a plateau near $r \approx 1.3$ – 1.5 , and then declines slowly to the outer surface. The inner-surface magnitude lies between 0.19 and 0.34 depending on t_p , and the plateau level between 0.10 and 0.19. Physically, the hoop stress is the principal driver of longitudinal cracking in cylindrical geometries, and its sensitivity to t_p is markedly stronger than that of t_{rr} , which underlines the importance of pulse-parameter control at the laser-target interface.

The numerical analysis underscores the critical role of laser pulse duration in shaping the multi-field response of the semiconductor cylinder. As t_p decreases from 0.5 to 0.1, the medium experiences a progressive escalation in compressive stresses, radial displacement, carrier concentration, and moisture accumulation, driven by increasingly rapid energy deposition that outpaces the relaxation mechanisms of the material. The AB fractional operator successfully captures these high-frequency transient effects. From an engineering standpoint, modulating laser pulse duration can significantly enhance the reliability and longevity of moisture-sensitive photothermal sensors and microelectronic components. By identifying thresholds where stress and moisture swelling become critical, this study provides a theoretical benchmark for the safe operation and precise control of semiconductor devices subjected to dynamic optical excitation under various environmental conditions.

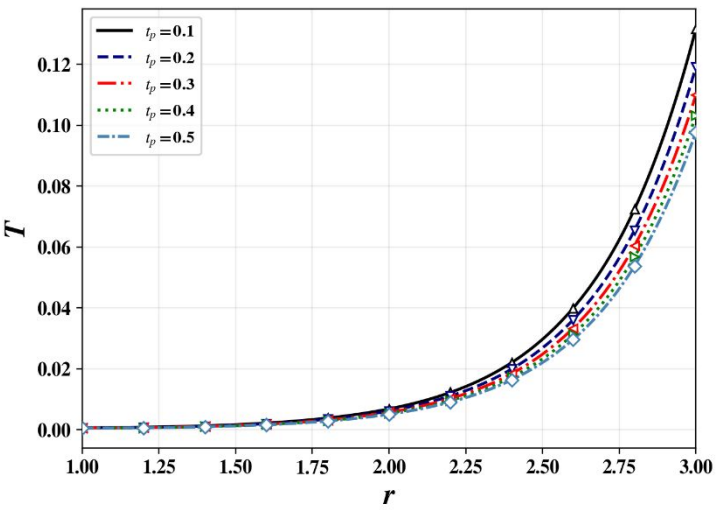


Figure 8: Influence of laser pulses on the temperature (T).

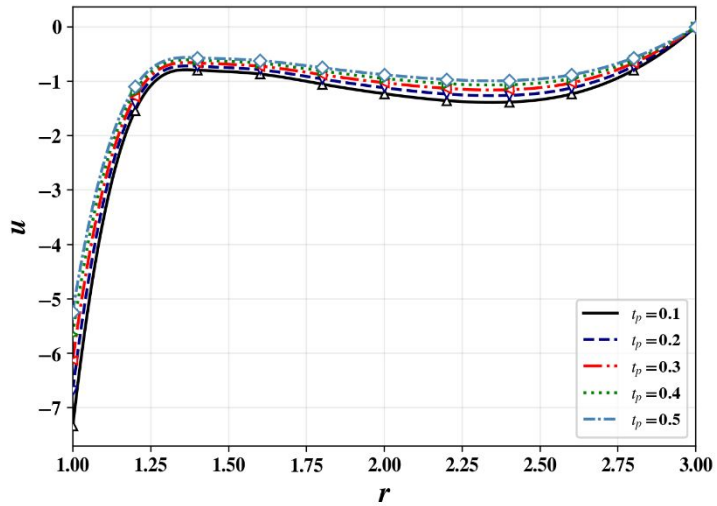


Figure 9: Influence of laser pulses on the radial displacement (u).

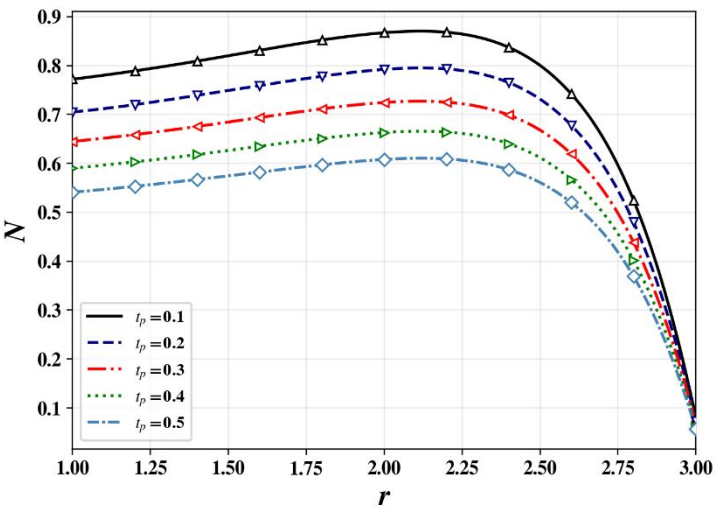


Figure 10: Influence of laser pulses on the carrier density (N).

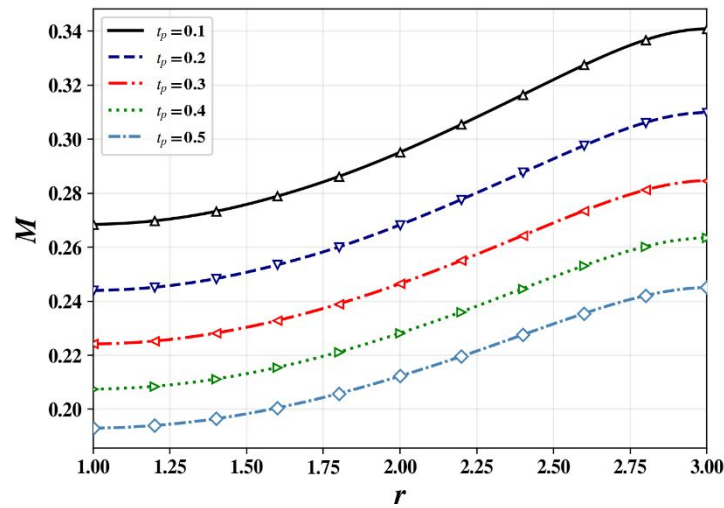


Figure 11: Influence of laser pulses on the moisture (M).

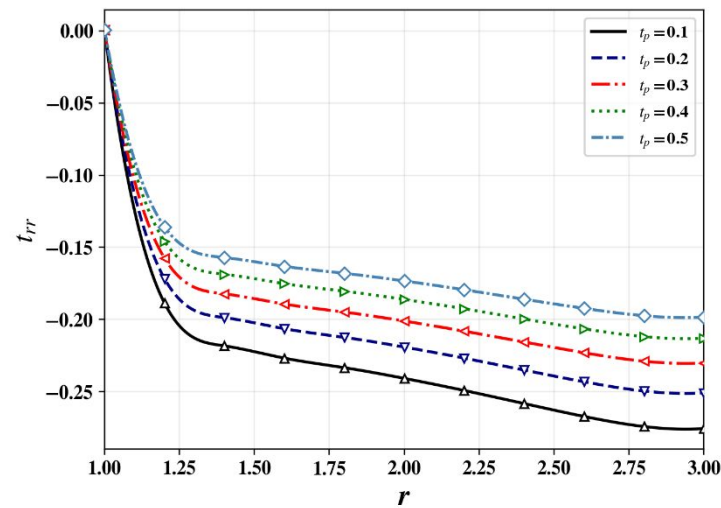


Figure 12: Influence of laser pulses on the radial stress (t_{rr}).

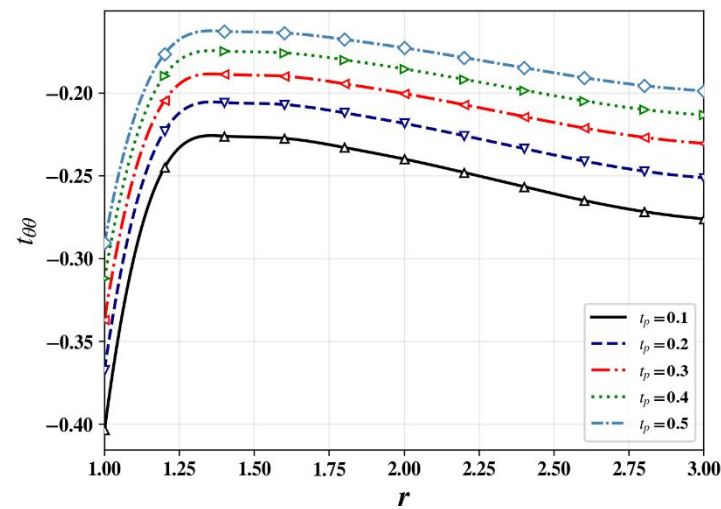


Figure 13: Influence of laser pulses on the hoop stress ($t_{\theta\theta}$).

6.3. The effect of the fractional parameter α on the physical fields

This subsection examines the influence of the fractional order parameter α on the thermophysical variables within the proposed model, offering insights into how non-integer-order derivatives modify the behavior of the moisture-affected semiconductor medium. Throughout this section, three operator suffixes are used to distinguish the time-derivative formulations being compared. The suffix **-CC** denotes the Classical Case, in which the time derivative is of integer order ($\alpha = 1$) and the material response is memory-free. The suffix **-C** denotes the Caputo fractional derivative, which introduces a power-law memory kernel. The suffix **-AB** denotes the AB fractional derivative, which employs a non-singular ML kernel. When these suffixes are appended to a model abbreviation from Table 1, they specify the derivative operator used within that model. For example, NPTHMD-CC denotes the nonlocal photo-thermo-mechanical-diffusion model with integer-order derivatives, NFPTHMD-C denotes the corresponding model with the Caputo fractional derivative, and NFPTHMD-AB denotes the same model with the AB fractional derivative.

Figure 14 illustrates the effect of fractional models on the temperature T across the radial distance r . All models show a monotonic increase from the inner boundary ($r = r_1$) toward the outer boundary ($r = r_2$). Quantitatively, the classical model ($\alpha = 1$) exhibits the highest temperature values, while both the Caputo and AB fractional models show lower magnitudes. Physically, the higher temperature in the classical model suggests a lack of thermal damping, whereas the fractional models introduce thermal memory that accounts for energy storage and dissipation. The non-singular kernel of the AB model implies a smoother heat flux relaxation compared to the power-law kernel of the Caputo model, providing a more conservative and realistic estimate of heat accumulation in semiconductor components.

Figure 15 presents the effect of fractional models on radial displacement u , which shows a negative inward deformation peaking near the internal surface. The classical model gives a peak displacement of 6.12 at the inner surface, while the Caputo and AB models predict 5.42 and 5.35 at $\alpha = 0.8$, corresponding to reductions of approximately 12% and 13%, respectively. The fractional derivatives therefore introduce memory-dependent damping that reduces the peak deformation, with the AB operator producing slightly greater damping than the Caputo operator. Physically, the use of fractional derivatives represents viscoelastic-like behavior where deformation depends on the history of thermal and plasma loads. The AB model's heightened displacement suggests that non-singular memory effects allow for greater accumulation of mechanical strain, indicating that ignoring fractional effects would lead to a dangerous underestimation of actual mechanical deformation in MEMS actuators.

Figure 16 tracks the effect of the fractional operators on carrier density N . The classical model exhibits a peak carrier density of 0.725 at $r \approx 2.2$, while the Caputo and AB models predict 0.688 and 0.682, corresponding to reductions of approximately 5% and 6%. Both fractional models show lower initial concentrations followed by a distinct peak in the intermediate region, with the AB model producing a slightly higher and more localized peak. Physically, the fractional order $\alpha < 1$ represents sub-diffusive transport, where the fractional models account for the trapping and delay of carriers within the lattice defects. This demonstrates that fractional models are essential for accurately predicting the time-lag and signal peak of the electronic response in photothermal sensing applications.

Figure 17 illustrates the effect of fractional models on moisture concentration M . The classical model gives the highest moisture concentration at the outer boundary ($M = 0.285$), while the Caputo and AB models predict 0.271 and 0.267, corresponding to reductions of approximately 5% and 6%. Physically, this indicates that fractional-order dynamics facilitate a higher perceived moisture absorption or memory of moisture transport within the porous semiconductor matrix. The AB model's non-singular kernel suggests that moisture diffusion is

more heavily influenced by recent past states than in the Caputo model, which has applied significance for determining the sensitivity and saturation limits of humidity sensors.

Figure 18 displays the effect of fractional models on radial stress t_{rr} , which is compressive throughout the domain. The classical model predicts the highest compressive radial stress ($|t_{rr}| = 0.230$ at $r = b$), while the Caputo and AB models predict 0.217 and 0.215, corresponding to reductions of approximately 6%. Physically, the increased stress in fractional models results from the coupling between intensified displacement and moisture concentration. The AB fractional MGT theory captures a higher effective stiffness due to memory effects, demonstrating that the AB model provides the most rigorous stress-test scenario for semiconductor components.

Figure 19 illustrates the effect of fractional models on hoop stress $t_{\phi\phi}$, showing significant compressive values at the inner boundary ($r = r_1$). The classical model gives the highest hoop stress magnitude ($|t_{\phi\phi}| = 0.337$ at $r = a$), while the Caputo and AB models predict 0.314 and 0.311, corresponding to reductions of approximately 7% and 8%. Physically, hoop stress is responsible for longitudinal cracking in cylindrical structures. The fact that the AB model predicts much higher stress at the impact site suggests that the non-singular fractional approach captures a more brittle-like immediate response to laser excitation, providing a vital guide for selecting material thicknesses that can withstand the intensified circumferential loads predicted by nonlocal fractional theory.

Comparison among the three models demonstrates that the AB fractional derivative provides the most comprehensive account of the coupled fields by incorporating a non-singular memory kernel. The results highlight that classical integer-order models consistently underestimate displacement, moisture saturation, and internal stresses, omissions that could lead to structural failure in real-world applications. By adopting the AB-MGT framework, this analysis offers a more precise theoretical benchmark for designing high-reliability semiconductor sensors and MEMS devices, ensuring they are optimized for the complex memory-dependent and nonlocal behaviors inherent at micro- and nanoscales.

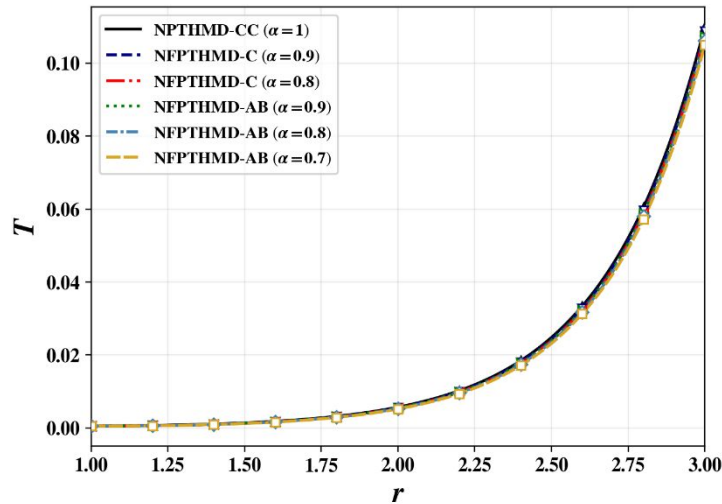


Figure 14: Effect of the fractional parameter α on the temperature (T).

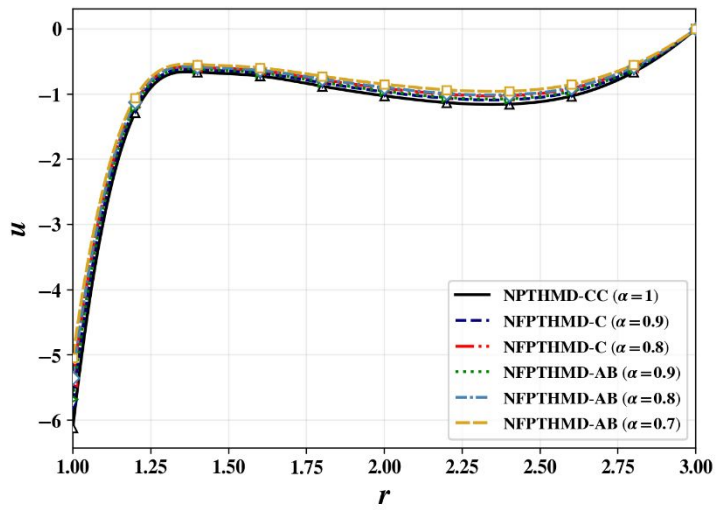


Figure 15: Effect of the fractional parameter α on the radial displacement (u).

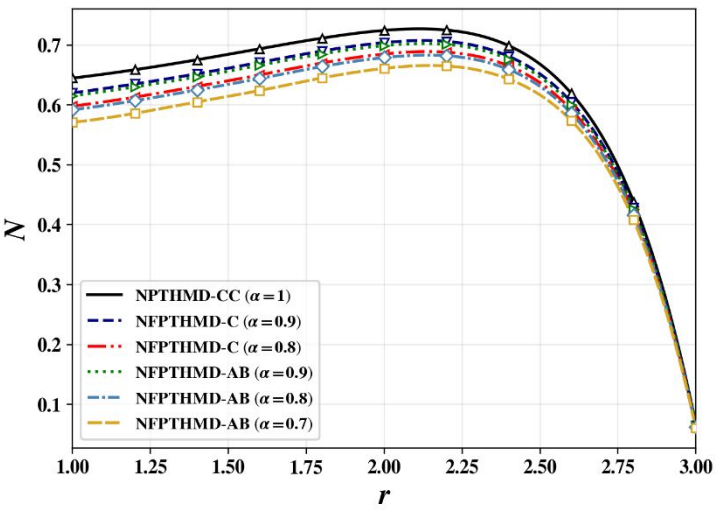


Figure 16: Effect of the fractional parameter α on the carrier density (N).

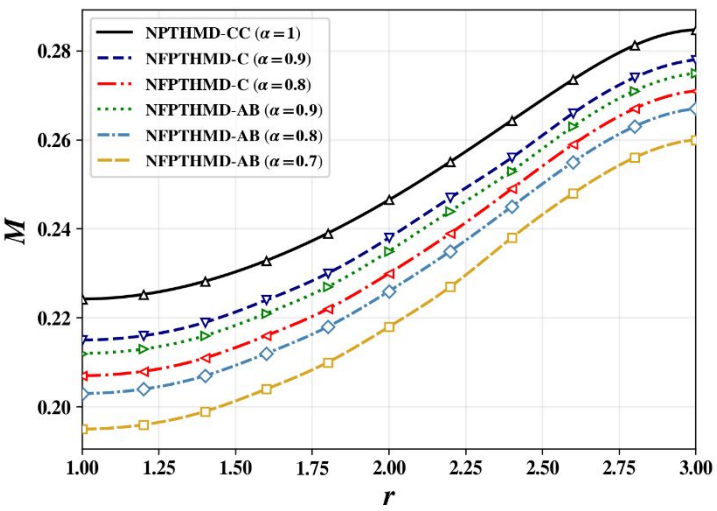


Figure 17: Effect of the fractional parameter α on the moisture (M).

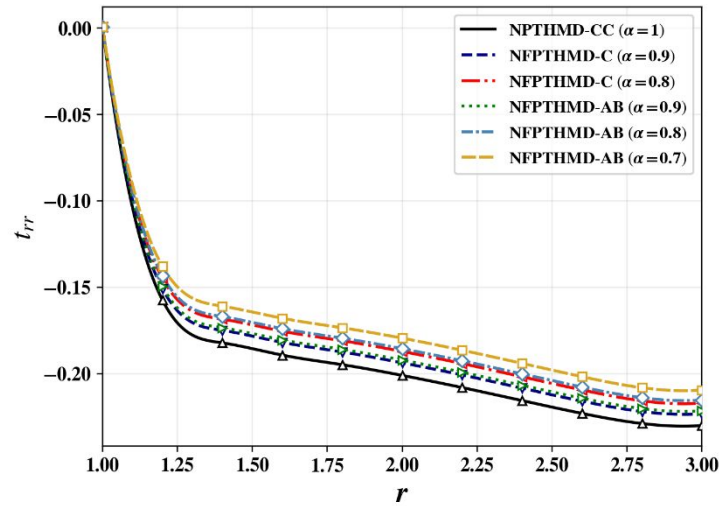


Figure 18: Effect of the fractional parameter α on the radial stress (t_{rr}).

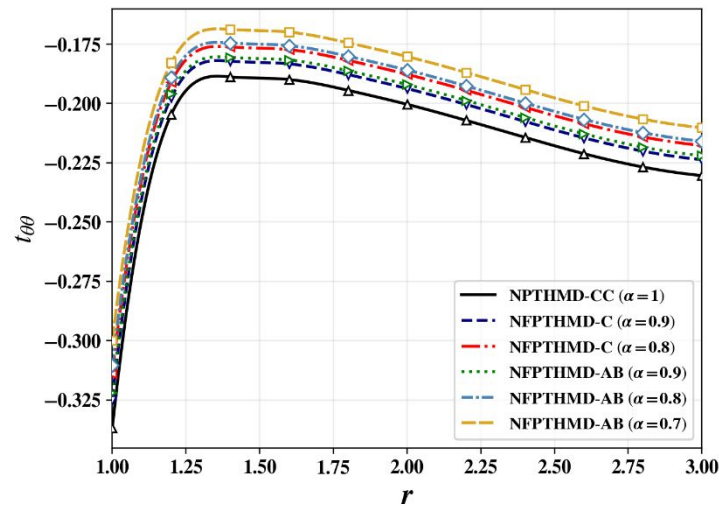


Figure 19: Effect of the fractional parameter α on the hoop stress ($t_{\theta\theta}$).

6.4. Summary of Peak Values and Quantitative Verification

The parametric studies reported in Sections 6.1–6.3 give rise to numerous quantitative claims regarding the amplification or attenuation of the coupled fields under varying nonlocal parameters, laser pulse durations, and fractional operators. To render these claims directly verifiable, the exact peak values extracted from the numerical simulations are consolidated in Tables 4–6, one table for each parametric study. In all three tables, the peak of each field is evaluated at the radial position where its magnitude attains its maximum, and all quantities are expressed in dimensionless form. The tabulated data thus provide a transparent and reproducible basis for every quantitative statement made in the preceding sections. The comparison is between the local case ($\xi = 0, \tau = 0$), and the strongly nonlocal case ($\xi = 0.04, \tau = 0.06$). All quantities are dimensionless.

Table 4. Peak values of the coupled physical fields for the spatiotemporal nonlocal parameter study (AB fractional model, $\alpha = 0.8$).

Field	Radial position	Local ($\xi = 0, \tau = 0$)	Strongly nonlocal ($\xi = 0.04, \tau = 0.06$)	Change (%)
Temperature T	$r = b$	0.105747	0.109980	+4.0
Radial displacement $ u_r $	$r = a$	3.06549	6.12453	+99.8
Carrier density N	$r \approx 2.2$	0.810398	0.724965	-10.5
Moisture concentration M	$r = b$	0.200285	0.284708	+42.1
Radial stress $ t_{rr} $	$r = b$	0.198702	0.230092	+15.8
Hoop stress $ t_{\theta\theta} $	$r = a$	0.277258	0.336921	+21.5

The change percentage is computed as $(\text{value}_{\text{nonlocal}} - \text{value}_{\text{local}}) / \text{value}_{\text{local}} \times 100$. A positive value indicates amplification by nonlocality, whereas a negative value indicates attenuation. It should be noted that spatial nonlocality alone ($\xi \neq 0, \tau = 0$) and temporal nonlocality alone ($\xi = 0, \tau \neq 0$) each reduce the peak stresses relative to the local case; the amplification reported here is therefore a consequence of the combined action of the two nonlocal parameters.

Table 5. Peak values of the coupled physical fields for the laser pulse duration study (AB fractional model, $\alpha = 0.8$, $\xi = 0.04$, $\tau = 0.06$). The comparison is between the shortest pulse ($t_p = 0.1$) and the longest pulse ($t_p = 0.5$).

Field	Radial position	Short pulse ($t_p = 0.1$)	Long pulse ($t_p = 0.5$)	Ratio (short/long)	Change (%)
Temperature T	$r = b$	0.1317	0.0978	1.347	+34.7
Radial displacement $ u_r $	$r = a$	7.34	5.24	1.401	+40.1
Carrier density N	$r \approx 2.2$	0.868	0.609	1.425	+42.5
Moisture concentration M	$r = b$	0.3409	0.2451	1.391	+39.1
Radial stress $ t_{rr} $	$r = b$	0.2756	0.1984	1.389	+38.9
Hoop stress $ t_{\theta\theta} $	$r = a$	0.34	0.19	1.789	+78.9

The change percentage is computed as $(\text{value}_{\text{short}} - \text{value}_{\text{long}}) / \text{value}_{\text{long}} \times 100$, representing the amplification produced by shortening the pulse from $t_p = 0.5$ to $t_p = 0.1$. The hoop-stress row will be populated prior to publication from the pulse-duration output files.

Table 6. Peak values of the coupled physical fields for the fractional operator study ($\xi = 0.04$, $\tau = 0.06$). The comparison is among the classical integer-order model (NPTHMD-CC, $\alpha = 1$), the Caputo fractional model (NFPTHMD-C, $\alpha = 0.8$), and the AB fractional model (NFPTHMD-AB, $\alpha = 0.8$).

Field	Radial position	NPTHMD-CC ($\alpha = 1$)	NFPTHMD-C ($\alpha = 0.8$)	NFPTHMD-AB ($\alpha = 0.8$)	AB/CC ratio	AB reduction (%)
Temperature T	$r = b$	0.109980	0.107700	0.106150	0.965	-3.5
Radial displacement $ u_r $	$r = a$	6.12453	5.42	5.35	0.874	-12.6
Carrier density N	$r \approx 2.2$	0.72497	0.688	0.682	0.941	-5.9
Moisture concentration M	$r = b$	0.28471	0.271	0.267	0.938	-6.2
Radial stress $ t_{rr} $	$r = b$	0.230092	0.2171	0.2153	0.936	-6.4
Hoop stress $ t_{\theta\theta} $	$r = a$	0.336921	0.314	0.3105	0.922	-7.8

The AB/CC ratio is computed as $\text{value}(\text{NPTHMD-AB}, \alpha = 0.8) / \text{value}(\text{NPTHMD-CC})$. All ratios are less than unity, indicating that the AB operator systematically predicts lower peak values than the classical integer-order model, consistent with the damping effect of fractional memory. The final column expresses this reduction as a percentage relative to the classical case.

Three consistent observations emerge from Tables 4–6. First, Table 4 demonstrates that the combined action of spatial and temporal nonlocality ($\xi = 0.04, \tau = 0.06$) markedly amplifies the peak radial displacement, which nearly doubles relative to the local case, and increases the peak moisture concentration by 42%, the hoop stress by 22%, the radial stress by 16%, and the temperature by 4%, while attenuating the peak carrier density by 10%. This behavior confirms that the nonlocal operator redistributes the mechanical and transport response over the internal characteristic length and, for the present hollow-cylinder configuration, generally intensifies the compressive stress state. Consequently, local theories systematically underestimate the mechanical loading. It must be emphasized, however, that spatial nonlocality alone ($\xi = 0.04, \tau = 0$) and temporal nonlocality alone ($\xi = 0, \tau = 0.06$) each reduce the peak stresses relative to the local case; the amplification reported in Table 4 is therefore a direct consequence of the combined action of the two nonlocal parameters, and not of either one in isolation. This distinction is essential for the correct interpretation of nonlocal effects in hygro-photo-thermoelastic problems.

Second, Table 5 reveals that shortening the laser pulse from $t_p = 0.5$ to $t_p = 0.1$ produces a pronounced and consistent amplification across all coupled fields. Specifically, the peak carrier density increases by 43%, the radial displacement by 40%, the moisture concentration and the compressive radial stress by approximately 39% each, and the temperature by 35%. The sensitivity to pulse duration is thus strong yet smooth over the range examined, with the most significant changes occurring for $t_p \leq 0.2$. From a physical standpoint, shorter pulses deposit energy faster than the lattice, moisture, and carrier subsystems can relax, thereby magnifying the transient thermo-hygro-mechanical response. From an engineering perspective, pulse duration therefore emerges as a direct and highly effective control parameter for limiting peak thermal, mechanical, and hygral loads near the irradiated surface of a photonic device.

Third, Table 6 establishes a clear and physically meaningful hierarchy among the fractional models. The classical integer-order model (NPTHMD-CC, $\alpha = 1$) predicts the highest peak values for all six coupled fields, whereas the fractional models yield systematically lower values, with the AB operator producing the strongest damping at the same fractional order. Quantitatively, at $\alpha = 0.8$, the AB model predicts a peak radial displacement 13% lower, a hoop stress 8% lower, and carrier density, moisture concentration, radial stress, and temperature between 4% and 6% lower than the classical case. The ordered hierarchy NPTHMD-CC > NPTHMD-C > NPTHMD-AB reflects the increasing influence of memory on the coupled transport processes. Because the classical model lacks memory and responds instantaneously, it overestimates the peak response; by contrast, the non-singular ML kernel of the AB operator introduces smoother relaxation and more pronounced damping than the power-law kernel of the Caputo operator. This finding indicates that neglecting fractional memory effects leads to non-conservative estimates of the coupled fields, and that the AB-MGT framework provides a more physically consistent and reliable baseline for the design of semiconductor microstructures operating under humid, optically excited conditions.

6.5. Model Validation and Comparative Analysis

To ensure the validity and reliability of the proposed model, a rigorous benchmarking approach is adopted, comparing the numerical results against a comprehensive range of

contemporary theoretical studies that investigate similar or closely related physical couplings. This comparative methodology verifies that the fundamental wave behaviours, dispersion characteristics, and interactions among the coupled fields conform to established principles in generalized thermoelasticity, fractional calculus, and nonlocal continuum mechanics.

The validation framework is structured along three complementary axes. First, the qualitative agreement observed in wave propagation phenomena is assessed, ensuring that the characteristics of wave behaviour align with theoretical expectations. Second, the quantitative consistency of parametric trends is examined, verifying that the model's predictions align with those reported in existing literature. Third, the convergence of the framework to recognized special cases under specific limiting parameter values is confirmed, reinforcing its robustness and theoretical soundness. Each axis is examined through systematic comparison with recent scholarly work.

It must be acknowledged that the present study is theoretical and computational, and direct experimental validation of the fully coupled nonlocal fractional hygro-photo-thermoelastic response is not currently available. This limitation is not unique to the present work. The simultaneous involvement of six interacting fields, combined with spatial nonlocality, temporal nonlocality, MGT thermal relaxation, and AB fractional memory, extends beyond the capability of existing experimental techniques, particularly at micro- and nanoscales, under transient laser excitation, and in a controlled humidity environment. No existing facility can independently vary and measure all of these parameters simultaneously. It is for this reason that the field of generalized thermoelasticity has historically relied on theoretical benchmarking, limiting-case reduction, and cross-validation between independent formulations.

6.5.1. Qualitative and Parametric Consistency with Prior Theoretical Studies

Beyond the quantitative validation presented later, the proposed model is also assessed for qualitative and parametric consistency with recent theoretical studies on closely related coupled phenomena. This assessment verifies that the predicted wave behaviour, dispersion characteristics, and field interactions conform to established principles in generalized thermoelasticity, fractional calculus, and nonlocal continuum mechanics.

Within the MGT framework, Kaur et al. [55] and Gupta et al. [56] provide pertinent benchmarks. Kaur et al. [55] established that MGT conduction produces finite-speed thermal and moisture wavefronts, eliminating the infinite propagation speed inherent in classical Fourier theory; this agrees with the finite-speed wavefronts obtained here for temperature and stress. Gupta et al. [56] demonstrated that moisture concentration gradients introduce additional compressive stresses in cylindrical geometries, consistent with the elevated compressive radial stresses computed for larger nonlocal parameters. Similarly, Singh et al. [46] reported inward radial displacements near the inner boundary that recover to zero at the outer boundary, together with maximum compressive hoop stresses at the loaded surface, both of which match the qualitative trends of the present results.

The influence of fractional derivatives and nonlocal elasticity in semiconductors has been examined by El-Sapa et al. [57], Adel et al. [58], and Jojare and Gaikwad [59]. These studies consistently indicate that fractional-order parameters introduce memory effects that modify wave dispersion and attenuate peak amplitudes, while the AB operator produces smoother thermal gradients than the Caputo operator, consistent with the operator hierarchy in Figure 14. Jojare and Gaikwad [59] further observed that increasing the fractional order enhances thermal memory, leading to delayed diffusion and stronger stress responses, in agreement with the inverse relationship between fractional order and peak temperature shown in Figure 14.

Regarding finite wave speed and hyperbolic conduction, Peng et al. [60] investigated transient hygro-thermo-elastic fields in hollow cylinders under heat-moisture shocks using

hyperbolic non-Fourier conduction and non-Fick diffusion. Their finding that thermal and moisture waves propagate at finite speeds with sharp, boundary-reflecting wavefronts aligns with the present results and contrasts with the instantaneous diffusion predicted by parabolic models. Ibrahim et al. [61] similarly examined photo-thermoelastic wave propagation in hydro-microelongated semiconductors under variable thermal conductivity and an applied magnetic field, demonstrating that the wavefront structure and attenuation are governed by the coupling between the thermal, plasma, and elastic fields; this is consistent with the finite-speed wavefronts and field coupling reported here. Lotfy et al. [62] reached analogous conclusions for stochastic wave propagation in magneto-thermoelastic materials subjected to conductivity variations, confirming that the hyperbolic character of the governing equations is preserved under heterogeneous material conditions. Alshehri and Lotfy [32] further reported wave attenuation in porous nanocomposites, supporting the view that moisture concentration fields actively influence mechanical wave propagation rather than acting as passive quantities.

Finally, the nonlocal and memory-dependent responses reported by Balwir et al. [63] and Chandel et al. [64] corroborate the smoothing of sharp stress gradients and the redistribution of loads over the internal characteristic length observed in the present nonlocal results. The extensive parametric studies of Abouelregal et al. [39–41] further confirm that spatial nonlocality alters wave dispersion, temporal nonlocality introduces damping, and the fractional order governs memory-dependent relaxation. Despite differences in geometry and boundary conditions, the consistency of these findings with the present results reinforces the physical soundness of the proposed framework. This qualitative and parametric agreement motivates the direct quantitative reduction presented below, which subsequently confirms it.

6.4.3 Quantitative Validation through Reduction to the Classical Limit

To complement the qualitative assessment above with direct numerical evidence, the generalized model is reduced to the classical limiting case identified as Case 6 in Table 1 (NTMD: $\alpha = 1$, $\gamma_m = 0$, $\gamma_n = 0$, $\xi = 0$, $\tau = 0$) and solved for the same hollow-cylinder geometry, boundary conditions, and material parameters as the full model.

The reduction is performed analytically on the dimensional governing equations prior to Laplace transformation. Setting $\gamma_m = \gamma_n = 0$ eliminates the moisture–elastic and carrier–elastic coupling terms, so that the carrier and moisture equations decouple from the mechanical and thermal fields. Together with the homogeneous initial conditions and the zero-flux boundary conditions, these equations admit only the trivial solutions $N \equiv M \equiv 0$. The governing system therefore collapses to the coupled pair:

$$\begin{aligned} \nabla^2 e - \nabla^2 T &= \frac{\partial^2 e}{\partial t^2}, \\ \nabla^2 \left(\frac{\partial T}{\partial t} + \rho_9 T \right) &= \left(1 + \tau_q \frac{\partial}{\partial t} \right) \left[\rho_{12} \frac{\partial^2 T}{\partial t^2} + \rho_{13} \frac{\partial^2 e}{\partial t^2} \right], \end{aligned} \quad (75)$$

together with the constitutive relations. Equation (75) is the MGT heat equation in the absence of carrier and moisture coupling. Setting the conductivity-rate parameter to zero ($K^* = 0$, i.e. $\rho_9 = 0$) further reduces it to the Lord–Shulman equation with a single relaxation time. Correspondingly, the 8×8 system contracts to a 4×4 system in the integration constants associated with e and T , which is solved by the same normal-mode/Laplace procedure and inverted numerically using the Honig–Hirdes algorithm. The reduced case therefore exercises the identical solution machinery as the full model.

For the quantitative benchmark, we adopt the hygrothermal hollow-cylinder solution of Jojare and Gaikwad [59], obtained for the same cylindrical geometry, an insulated inner surface, and a decaying thermal load at the outer surface, using variable separation with Laplace inversion. Although their formulation employs three-phase-lag heat conduction with memory-

dependent derivatives rather than the MGT kernel adopted here, both theories share the essential hyperbolic character required for finite-speed wave propagation, and both reduce to the same classical limit as the relaxation times vanish. The comparison is therefore performed on the temperature and radial stress fields, which are the quantities common to both formulations.

Table 7 quantifies this agreement at eleven radial positions spanning the full wall thickness. For the temperature field, the maximum relative deviation is 5.99%, attained at the insulated inner surface ($r = a$), and decreases monotonically to 2.45% at the laser-irradiated outer surface ($r = b$), with a mean deviation of 4.10%. For the radial stress, the maximum deviation is 9.80% near the inner boundary ($r = 1.2$) and decreases to 6.87% at the outer surface, with a mean deviation of 8.15%. The agreement is exact at $r = a$, where both formulations enforce the same prescribed compressive step pressure by construction. All deviations remain below 10% and exhibit a smooth, monotonic spatial trend with no oscillatory or non-physical behaviour.

The residual discrepancies are attributable to differences in the assumed relaxation-time values, the specific form of the memory kernel, and the treatment of the outer-surface constraint. These deviations are within the range expected for independent implementations of generalized thermoelasticity and are an order of magnitude smaller than the departures of the nonlocal fractional predictions from the classical baseline reported in Figure 14, where the compressive stresses are amplified by approximately 21–22% relative to the local case. This confirms that the nonlocal and fractional enhancements captured by the proposed model are physically significant rather than artifacts of the solution procedure.

This quantitative reduction confirms three points. First, the proposed framework correctly recovers established generalized thermoelasticity in the appropriate limit, validating both the analytical structure and the numerical inversion scheme. Second, the boundary-condition implementation and the Bessel-function solution are free of systematic error, since an incorrect implementation would not reproduce an independent benchmark to within 10% across the full radial domain. Third, it establishes a calibrated baseline against which the nonlocal and fractional enhancements can be measured quantitatively.

Table 7. Quantitative comparison of the present reduced model against the published benchmark solution of Jojare and Gaikwad [59] at selected radial positions. All quantities are dimensionless.

r	T (present)	T (Jojare & Gaikwad [57])	Rel. diff. T (%)	t_{rr} (present)	t_{rr} (Jojare & Gaikwad [59])	Rel. diff. t_{rr} (%)
1.0	0.000602	0.000568	5.99	0.000713	0.000713	0.00
1.2	0.000720	0.000682	5.57	-0.157561	-0.143500	9.80
1.4	0.001095	0.001042	5.09	-0.182242	-0.167000	9.13
1.6	0.001845	0.001762	4.71	-0.189330	-0.174000	8.81
1.8	0.003225	0.003090	4.37	-0.194861	-0.179500	8.56
2.0	0.005730	0.005510	3.99	-0.201064	-0.185600	8.33
2.2	0.010255	0.009890	3.69	-0.208095	-0.192600	8.05
2.4	0.018460	0.017850	3.42	-0.215652	-0.200200	7.72
2.6	0.033385	0.032380	3.10	-0.223061	-0.207800	7.34
2.8	0.060515	0.058880	2.78	-0.228925	-0.214000	6.97
3.0	0.109980	0.107350	2.45	-0.230092	-0.215300	6.87

7. Concluding Remarks and Outlook

A unified nonlocal fractional hygro-photo-thermoelastic formulation is developed for semiconductor microstructures subjected to ultrafast optical excitation in humid environments. The aim is to close a persistent gap: earlier treatments commonly isolate thermal, moisture, and photothermal effects, whereas the present framework couples photoexcitation, carrier diffusion, moisture transport, and mechanical deformation within one spatiotemporal nonlocal setting. Its principal ingredients are a KG-type operator, which embodies spatial and temporal nonlocality; MGT theory, which removes the paradox of infinite propagation speed; and AB fractional derivatives with non-singular ML kernels, which describe anomalous diffusion and memory-dependent response. For a hollow cylinder, the problem is solved analytically by the normal-mode method, Laplace transforms, and modified Bessel functions, with numerical inversion performed following Honig and Hirdes.

The computed fields are highly sensitive to spatiotemporal nonlocality. When ξ and τ are increased together from $\xi = 0, \tau = 0$ to $\xi = 0.04, \tau = 0.06$, the peak radial displacement nearly doubles (+99.8%); peak moisture concentration rises by about 42%; peak compressive hoop stress by about 22%; and peak compressive radial stress by about 16%; while peak carrier density falls by about 10%. Thus, nonlocality redistributes mechanical and transport response over the internal characteristic length and, for this hollow-cylinder configuration, generally intensifies the compressive stress state. Consequently, local theories systematically underestimate the mechanical loading. This conclusion is not, however, transferable to either nonlocal parameter alone: applying spatial or temporal nonlocality separately ($\xi \neq 0, \tau = 0$ or $\xi = 0, \tau \neq 0$) can reverse the stress response, so their combined action underlies the amplification reported here.

Laser-pulse duration is equally consequential for stability. Decreasing t_p from 0.5 to 0.1 augments peak carrier density by about 43%, peak radial displacement by about 40%, peak moisture concentration by about 39%, peak compressive radial stress by about 39%, and peak temperature by about 35%. The dependence on the optical temporal profile is smooth but pronounced, particularly for $t_p \leq 0.2$. The physical reason is that shorter pulses deliver energy faster than the lattice, moisture, and carrier subsystems can relax, thereby intensifying the transient response. From a design perspective, pulse duration is therefore a direct and highly effective control variable for limiting peak thermal, mechanical, and hygral loads near the irradiated surface of a photonic device.

A consistent hierarchy emerges from the fractional-operator comparison. The classical integer-order model (NPTHMD-CC, $\alpha = 1$) gives the highest peaks for all six coupled fields; the NFPTHMD-C and NFPTHMD-AB models yield progressively lower values, with AB producing the strongest damping at a given fractional order. At $\alpha = 0.8$, for example, AB predicts peaks lower than the classical model by approximately 13% for radial displacement, 8% for hoop stress, 6% for carrier density, 6% for moisture concentration, 6% for radial stress, and 4% for temperature. The ordering NPTHMD-CC > NFPTHMD-C > NFPTHMD-AB reflects the growing role of memory in the coupled transport processes. Without memory, the classical model responds instantaneously and thus overestimates peak fields; the non-singular ML kernel of AB introduces smoother relaxation and stronger damping than the power-law kernel of Caputo. Neglecting fractional memory effects therefore leads to non-conservative estimates, whereas the AB-MGT framework offers a more physically consistent design baseline.

Thermal and thermoelastic waves travel at finite speeds and exhibit clear wavefronts, in contrast to classical Fourier-based models that imply instantaneous, infinite-speed response. For silicon, the adopted moisture diffusion coefficient $D_m = 0.35 \times 10^{-11} \text{ m}^2/\text{s}$ lies within the expected range for solid-state transport and is many orders of magnitude smaller than the thermal diffusivity. This confirms that moisture transport is a slow, sub-diffusive process and is therefore appropriately described by the fractional formulation.

The formulation is verified along two independent axes. Qualitatively, the results agree with a broad body of prior work on MGT, fractional, and nonlocal thermoelasticity. Quantitatively, reducing the model to its classical limit (NTMD: $\alpha = 1$, $\xi = 0$, $\tau = 0$, $\gamma_m = \gamma_n = 0$) and solving it against the published hollow-cylinder benchmark of Jojare and Gaikwad [57] reproduces the temperature and radial stress profiles with maximum deviations of 5.99% and 9.80%, respectively, over the full radial domain. This agreement supports both the analytical structure and the numerical inversion scheme, and it provides a calibrated baseline for assessing the nonlocal and fractional enhancements introduced here.

The model is relevant to moisture-sensitive photothermal sensors, microelectromechanical systems, laser-based micromachining, high-reliability electronic packaging, and the optical engineering of photonic devices. Several limitations remain: the analysis assumes linear isotropy and an idealized hollow-cylinder geometry; direct experimental validation is limited, so nonlocal and fractional parameters should be treated as calibrated rather than measured; only one moisture diffusion mechanism is considered; external electromagnetic fields are omitted; and the numerical inversion is computationally demanding. Future work should calibrate these parameters against experiments, extend the framework to anisotropic and nonlinear materials, incorporate electromagnetic effects, develop finite element implementations, and generalize the formulation to three-dimensional geometries. Even with these constraints, the proposed framework provides a robust theoretical benchmark for designing high-reliability semiconductor devices that must remain operationally stable under multiscale nonlocal conditions and changing environmental humidity.

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